Numerical Experiments for Lévy Score-Corrected Compound-Poisson Equilibrium Sampling

Abstract

We evaluate Lévy score-corrected compound-Poisson (LSC-CP) sampling on four multimodal Boltzmann model systems: a one-dimensional double well, a planar forty-mode Gaussian mixture, a depth-balanced three-well Müller–Brown landscape embedded in ten dimensions, and a coupled ϕ^4 chain in twenty-four dimensions. Across the reported horizons, LSC-CP markedly reduces terminal basin-population error relative to local samplers and, in E2–E4, relative to raw compound-Poisson dynamics; it also yields nonzero post-settling effective sample sizes for every basin indicator. E1 illustrates why basin occupancy alone is insufficient: raw CP happens to recover the two basin masses while still distorting tail and energy statistics. With the current MoG40 jump interval $[10, 14]$, its sliced W_2 lies within the finite-sample reference band, whereas its MMD remains marginally above that band. Thus, agreement of basin populations or one projection-based metric does not imply agreement of the complete distribution. The implementation uses finite chord quadrature, tamed Euler updates, an exponent cap, and projection onto a finite computational box. These practical stabilizations control overflow and large updates and improve numerical robustness in the stiff score regime, but they define an approximate finite-step kernel. Potential-energy discrepancies that are large relative to the chain-mean MCSE, conditional on the frozen reference estimate, are therefore reported explicitly. The evidence supports accelerated interbasin exploration and accurate population recovery with observable-dependent numerical bias, rather than exact finite-step Boltzmann sampling. No physical kinetics is inferred.

1 Overview and compared methods

All experiments target a Boltzmann law

$$\pi_\beta(x) = Z_\beta^{-1} e^{-\beta V(x)}, \qquad x \in \mathbb{R}^d. \tag{1}$$

The sampler of interest augments overdamped Langevin dynamics with a finite compound-Poisson jump process of measure $\nu(\,\mathrm{d}r) = \lambda\rho(\,\mathrm{d}r)$ and adds the stationary Lévy-score drift $S_{\nu,\beta}$ so that the combined generator

$$A_{\nu,\beta} f = \left[-\nabla V + S_{\nu,\beta}\right] \cdot \nabla f + \beta^{-1}\Delta f + J_\nu f \tag{2}$$

is weakly stationary for π_β (Appendix A). The practical estimators evaluate the score integral by deterministic chord quadrature where affordable and by *paired* random-bank estimators in higher dimension: the realized jump displacements and the Lévy-score drift of one step are generated from the same random finite measure. This pairing aligns the drift and jump measures before the remaining finite chord quadrature is applied (Appendix A.1). The reported variants are: deterministic shell quadrature (double well), full numerical chord–radius–angle quadrature for the annular MoG40 law, the paired single-atom estimator LSC-CP-RA (double well and MoG40, as an estimator-agreement check), deterministic shell quadrature in the two higher-dimensional systems, and the paired multi-atom estimator LSC-CP-MA (Müller–Brown and ϕ^4). The numerical artifacts store the latter under the key `LSC-CP-MA`; legacy plotting aliases call it `LSC-CP-RA (4)` or `LSC-CP-RA (8)`. We use MA throughout this note because the implementation draws one displacement from every atom family, distinguishing it from the single-displacement RA estimator in E1–E2.

Baselines are the unadjusted Langevin algorithm (ULA), Metropolis adjusted Langevin (MALA), the BAOAB underdamped integrator (BAOAB), an α-stable Lévy-flight analogue (FLA), and parallel tempering (PT) with an automatically tuned ladder and fully counted replica cost. Raw compound Poisson (CP), i.e. the same jump bank *without* the score correction, is included as a stationarity-defect ablation: its invariant law is not π_β, so it is a control for the value of the correction, not a fair efficiency baseline. All methods share per-seed initial ensembles. In the double-well and MoG40 full-law experiments, CP and deterministic/numerical-quadrature LSC-CP additionally share the jump random stream pathwise. In the Müller–Brown and ϕ^4 experiments, raw full-law CP and LSC-CP-MA match the marginal jump geometry, weights, and intensity, but do not use the same realized random banks and hence are not pathwise paired.

The implemented scheme uses finite chord quadrature, log-domain accumulation with an exponent cap, tamed Euler drift updates, and projection onto a finite computational domain to stabilize evaluation in the stiff-score regime. These operations define the finite-step algorithm evaluated below. The stationarity identity applies to the ideal continuous-time generator with the exact, uncapped Lévy-score field on an unbounded state space. We assess the equilibrium fidelity of the discretization through timestep and quadrature sensitivity checks, observable biases, and cap/projection activation rates. Passage statistics are interpreted solely as algorithmic mixing diagnostics; no physical reaction kinetics is inferred.

2 Benchmark systems

	Double well	MoG40	Depth-balanced MB	Coupled ϕ^4
dimension d	1	2	10	24
β	8	8	24	8
particles N	16,000	2,500	2,000	1,000
horizon T	100	100	200	100
Δt (used)	0.005	0.010	0.005	0.002
steps	20,000	10,000	40,000	50,000
seeds	24	24	16	16
jump rate λ	1	1	1	1
score quadrature	q_θ=16, q_ρ=4	q_θ=16, q_ρ=4, m_ϕ=16	q_θ=8, q_ρ=4	q_θ=32, q_ρ=4
score estimator	deterministic + RA	numerical annulus + RA	deterministic + MA	deterministic + MA
weak-identity residual	1.5×10^{-7}	2.1×10^{-8}	5.0×10^{-7}	2.0×10^{-7}

Table 1: Reported numerical configurations and uncapped weak drift–jump identity residuals (diagnostic tolerance 10^{-6}; Appendix E). The residual measures generator-level drift–jump cancellation for a finite family of test functions. Here N is the number of particles per seed block; a batched run with S seeds evolves SN particles and computes metrics separately on each N-particle block. Full model, bank, and ladder specifications are in Appendix B.

Artifact–source distinction for MoG40. Table 1 records the configuration that generated the saved results: $[10, 14]$ and $m_\phi = 16$. The current experiment builder retains $[10, 14]$ but raises the default angular order to $m_\phi = 64$ for a future rerun. Because no sampler was rerun for this revision, all numerical tables remain aligned with the saved $m_\phi = 16$ artifacts and must not be attributed to the newer default. The saved E2 manifest does not serialize the annulus endpoints or resolved estimator. Its recorded repository SHA resolves to the older $[4, 15]$ law with analytic `MoG40Score`, whereas the jump-term certificate ratios and evaluation ledger identify the executed configuration as $[10, 14]$ with numerical `ShellScore` at $m_\phi = 16$. Thus the executed law and estimator are identifiable from numerical fingerprints, but the exact executed source-tree provenance is not recoverable from the archived metadata. A future production run should serialize both the requested and resolved configurations.

Double well ($d = 1$). $V(x) = (x^2 - 1)^2$ at $\beta = 8$: two equal wells at ± 1 separated by a dimensionless barrier $\beta \Delta V = 8$. The jump bank is a symmetric shell at ± 2 with radial half-width 0.2. This is the transparent two-state test of target recovery and interwell communication. Every main relaxation run starts from

$$X_0 = -1 + 0.05\,\xi, \qquad \xi \sim N(0, 1). \tag{3}$$

MoG40 ($d = 2$). Using NumPy's PCG64 generator with seed zero, the centers are drawn once as $\mu_k \sim \mathrm{Unif}([-40, 40]^2)$ and then frozen (the code call is `default_rng(0)`). The target and potential are

$$\pi_\beta = \frac{1}{40} \sum_{k=0}^{39} N(\mu_k, I_2), \qquad V(x) = -\beta^{-1} \log \sum_{k=0}^{39} e^{-\|x-\mu_k\|^2/2}. \tag{4}$$

The main relaxation ensemble is $X_0 = \mu_0 + 0.5\,\xi$, $\xi \sim N(0, I_2)$, and basin labels are assigned by the nearest center. The jump law is deliberately generic: radius uniform on $[10, 14]$ and direction uniform on the circle. The interval is narrower than the former $[4, 15]$ rule, while its upper radius retains a connected mode graph under both recorded mode-hit thresholds. Neither the annulus law nor `ShellScore` receives the mode list as a separate geometry-design input; the numerical score accesses the centers only through black-box evaluations of V. Exact-reference sampling, nearest-center labels, and the connectivity check use the centers directly. This is the many-basin stress test.

Depth-balanced Müller–Brown ($d = 10$). A Müller–Brown-type three-well landscape whose well depths are retuned to be equal, evaluated in a two-dimensional latent plane and embedded in ten dimensions by a fixed linear map with Gaussian auxiliary coordinates ($\sigma_{\mathrm{aux}} = 0.4$), at $\beta = 24$. The equal depths separate multimodality from metastability: the channels retain dimensionless barriers $\beta b_{AB} \simeq 11.1$ and $\beta b_{BC} \simeq 4.0$ while the equilibrium masses stay comparable. The bank has four directed relay atoms $A \leftrightarrow B \leftrightarrow C$ (no direct A–C atom), so transport must use the relay through B. This tests curved, nonuniform basins and a designed jump graph. If z_C is the refined latent minimum of well C, the main initial law is

$$z_0 = (z_C, 0_8) + 0.05\,\xi, \qquad X_0 = z_0 B^\mathsf{T}, \qquad \xi \sim N(0, I_{10}). \tag{5}$$

Coupled ϕ^4 chain ($d = 24$). Twelve periodic two-component sites, $V(q) = \frac{\kappa}{2\delta} \sum_i \|q_{i+1} - q_i\|^2 + \delta \sum_i W(q_i)$ with $\kappa = 2.5$, $\delta = 1/12$, $\beta = 8$, and a tilted double-double-well site potential W (Appendix B). The target has four coherent phases; the bank has eight directed edge atoms along the phase square (diagonals excluded because their coherent chords cross the field-zero hilltop). The primary collective variable is the spatial mean $\bar{q} = \frac{1}{12} \sum_i q_i \in \mathbb{R}^2$. This is the many-body example:

a collective transition in a coupled field. The main relaxation runs start from the coherent minus–minus phase,

$$Q_0 = m_{--} + 0.05\,\xi, \qquad \xi \sim N(0, I_{24}). \tag{6}$$

The four systems form a hierarchy of equilibrium questions—two-state balance, many-mode occupancy, curved-basin relay transport, and a collective field transition—rather than four independent leaderboard tasks.

3 Protocol, references, and numerical parameter selection

Run protocol. For each system all seeds are batched into one ensemble per method with per-seed initial blocks; every method starts from the identical localized per-seed ensemble. These labels denote blocks within one batched method run, not independently launched jobs; the dynamics use one method-level random generator, and batch wall time and NFE are repeated in the seed rows. Consequently, the reported seed standard deviation describes block-to-block variation within that batch rather than run-to-run timing uncertainty. We used dyadic timestep sensitivity checks based on terminal W_2, basin TV, MMD, and explored-mode-coverage diagnostics for the π_β-targeting methods. The non-targeting FLA and raw CP chains were recorded but did not determine the timestep. Score quadrature was selected analogously, with the additional uncapped weak-identity residual. These checks guided numerical parameter selection, but they did not include post-settling mean energy or collective-variable bias and therefore do not establish convergence of the invariant measure of the stabilized chain. The saved refinement helper selects the first method whose key begins with `LSC-CP`; in E3 and E4 this is the deterministic-quadrature arm. Accordingly, the quadrature-refinement and π-start hold gates do not separately validate the paired MA arm, which is assessed only through its terminal and post-settling diagnostics.

The explored-mode-coverage (EMC) diagnostic is higher-is-better, whereas the current automated agreement helper is written for lower-is-better errors. Consequently, EMC is recorded in the refinement tables but does not provide a valid acceptance gate in the saved runs. The refinement evidence should therefore be read as coming from W_2, basin TV, MMD, and the weak-identity residual. A corrected implementation should gate on $1 - \text{EMC}$.

References and finite-sample levels. The double well uses a dense-grid inverse-CDF reference; MoG40 an exact mixture draw; Müller–Brown a fine two-dimensional latent grid combined with exact Gaussian auxiliaries; ϕ^4 a Laplace-mixture proposal corrected by self-normalized importance weights (direct SNIS for basin masses, energy moments and the energy-overlap histogram, CV means, and the free-energy surface; finite sampling-importance-resampling only where unweighted clouds are required). The ϕ^4 weight diagnostics for the reference are: 2×10^5 proposals, weight ESS 41,182 (fraction 0.206), maximum normalized weight 3.6×10^{-3} (Appendix D). A separate $t = 300$ parallel-tempering check gave phase masses $(0.365, 0.204, 0.234, 0.197)$, whose TV distance from the SNIS masses is 0.042. Consequently, differences below this scale in the ϕ^4 basin metrics are reference sensitive. For W_2, MMD, basin TV, EMC, E1 density TV, and E3 ten-dimensional sliced W_2, a finite-sample reference level is estimated from 20 independent reference replicates at matched sample size. For lower-is-better errors, "within the reference band" means below mean $+\,3\,\text{sd}$. No such band is currently computed for FES RMSE, basin KL, energy-moment errors, energy-overlap deficit, KSD, or the archived first-CV CDF/KDE discrepancies; those quantities must not be described as being "at the floor."

Terminal-sample and reference-density figures. The method-sample panels were reconstructed from the saved `positions.csv` files; the reference-density panels were evaluated separately from the target density or reference integral described below. No sampler trajectory was rerun. Each double-well method is shown with the normalized target density $p(x) \propto \exp[-8(x^2-1)^2]$ and a deterministic display subset of its saved samples; the panel text contains neither density qualifiers nor point counts. For E2–E4, each method has one terminal scatter plot and each experiment has one separate reference-density plot. E2 evaluates the exact forty-component Gaussian-mixture density, and E3 evaluates $p(z) \propto \exp[-24V_{\mathrm{MB3}}(z)]$ in the active plane. The E4 visualization uses an independent, fixed-seed CPU direct-SNIS construction followed by a phase-conditioned weighted KDE of $\bar{q}$; it is not an analytic marginal and is distinct from the archived GPU SNIS cloud used for numerical metrics (Appendix D). Plot titles contain only the method name or "Reference."

Stability diagnostics. We monitored nonfinite states, score-cap and state-box projection frequencies, jump-boundary contact per applied jump, and basin-map-domain escape. The uncapped weak-identity residual was evaluated on a domain extending at least one maximum jump beyond the high-probability region. The measured values and domain-coverage rationale appear in Appendix E. These quantities diagnose numerical robustness and exposure to the stabilizations; small activation frequencies do not prove that the finite-step kernel has invariant law π_β.

4 Equilibrium recovery

Four observations organize Table 2 and Figure 1.

(i) Local dynamics remain strongly metastable. On every system, the terminal ULA, MALA, and BAOAB ensembles remain strongly localized relative to the target (basin TV 0.31–0.93, basin KL 1.1–4.2). Occasional label changes occur in some of the post-settling MoG40 and Müller–Brown traces, but they are too sparse to recover all basin populations. This quantifies the metastability that the nonlocal methods are intended to address.

(ii) LSC-CP improves basin recovery, with metric-dependent residual error. For the reported LSC estimator in each system, both the main W_2 and basin TV lie within their finite-sample reference bands. In MoG40, the numerical values are $W_2 = 2.05$ versus an upper reference level 2.142, and basin TV 0.053 versus 0.069. Its MMD, 0.03199, is slightly above the corresponding upper level 0.03171, so the full two-dimensional discrepancy has not been shown to be indistinguishable from reference sampling noise. Basin KL is small for the LSC estimator in every system, but no KL reference band was computed. In ϕ^4, differences in basin metrics below 0.042 remain sensitive to the independent-reference discrepancy described in Section 3.

The updated, uniformly weighted augmented-grid FES metric changes the earlier ordering. Parallel tempering has lower FES RMSE than LSC-CP on the double well (0.393 versus 1.210), MB (0.616 versus 1.053 for the multi-atom estimator), and ϕ^4 (1.063 versus 1.294). On MoG40 the corresponding values are 0.574 and 0.695. Thus LSC-CP improves the low-dimensional FES relative to local and raw-jump dynamics, but it is not the most accurate method under this FES definition. The α-stable flight can have a small value for an individual terminal metric, but its invariant law is not π_β and the number cannot be interpreted as evidence of correct equilibrium sampling.

(iii) The score correction substantially reduces the raw-jump distortion. In the double-well and MoG40 full-law runs, raw CP and LSC-CP share jump streams pathwise. The MB and ϕ^4 comparisons instead match the marginal bank geometry, weights, and intensity but are not pathwise paired. Raw CP retains basin TV 0.309/0.135/0.077 on MoG40/MB/ϕ^4. Its terminal mass outside

Method	W_2	$\mathrm{TV}_{\mathrm{basin}}$	$\mathrm{RMSE}_{F/(k_{\mathrm{B}}T)}$	$D_{\mathrm{KL}}(p^{\star}\|\widetilde{p})$	FES outside
Double well	(reference levels: W_2 0.061 ± 0.033, TV 0.0032 ± 0.0017)				
ULA	1.26 (0.003)	0.471 (0.001)	1.63 (0.06)	1.09 (0.02)	0
MALA	1.26 (0.003)	0.473 (0.001)	1.63 (0.04)	1.12 (0.02)	0
BAOAB	1.29 (0.002)	0.488 (0.001)	1.99 (0.07)	1.54 (0.04)	0
FLA	0.096 (0.018)	0.0033 (0.003)	1.32 (0.03)	3.4×10^{-5}	0.0033
PT	0.064 (0.032)	0.0033 (0.002)	0.393 (0.089)	3.2×10^{-5}	0
CP	0.176 (0.013)	0.0036 (0.003)	1.58 (0.02)	4.2×10^{-5}	0.029
LSC-CP	0.111 (0.017)	0.0036 (0.003)	1.21 (0.04)	4.4×10^{-5}	0.0047
LSC-CP-RA	0.131 (0.017)	0.0031 (0.003)	1.33 (0.06)	3.2×10^{-5}	0.0084
MoG40	(reference levels: W_2 1.31 ± 0.28, TV 0.051 ± 0.006)				
ULA	26.1 (0.02)	0.916 (0.003)	1.917 (0.008)	4.20 (0.02)	0
MALA	26.1 (0.03)	0.917 (0.003)	1.917 (0.010)	4.21 (0.014)	0
BAOAB	26.2 (0.03)	0.930 (0.004)	1.908 (0.007)	4.24 (0.016)	0
FLA	15.25 (0.21)	0.436 (0.007)	1.481 (0.057)	0.743 (0.045)	0.0011
PT	6.54 (0.34)	0.141 (0.007)	0.574 (0.034)	0.055 (0.005)	0
CP	11.34 (0.18)	0.309 (0.007)	1.761 (0.034)	0.319 (0.020)	0.316
LSC-CP	2.05 (0.29)	0.053 (0.006)	0.695 (0.043)	0.0089 (0.0020)	0.0023
LSC-CP-RA	2.59 (0.25)	0.111 (0.006)	0.888 (0.026)	0.0448 (0.0068)	0.0085
Depth-balanced MB, 10D	(reference levels: W_2 0.050 ± 0.020, TV 0.011 ± 0.007)				
ULA	0.409 (0.003)	0.307 (0.002)	1.961 (0.091)	1.17 (0.14)	0
MALA	0.411 (0.003)	0.308 (0.001)	1.966 (0.083)	1.22 (0.16)	0
BAOAB	0.415 (0.003)	0.309 (0.001)	2.095 (0.071)	1.52 (0.21)	0
FLA	0.179 (0.006)	0.119 (0.008)	1.743 (0.026)	0.048 (0.007)	0.039
PT	0.056 (0.021)	0.013 (0.008)	0.616 (0.036)	5.7×10^{-4}	0
CP	0.353 (0.008)	0.135 (0.006)	1.654 (0.024)	0.047 (0.005)	0.172
LSC-CP	0.050 (0.010)	0.017 (0.009)	1.070 (0.059)	0.00105 (0.00108)	0.0062
LSC-CP-MA	0.047 (0.009)	0.014 (0.006)	1.053 (0.053)	5.8×10^{-4}	0.0062
Coupled ϕ^4, 24D	(reference levels: W_2 0.130 ± 0.035, TV 0.024 ± 0.010)				
ULA	1.22 (0.004)	0.671 (0.003)	2.419 (0.082)	3.07 (0.31)	0
MALA	1.22 (0.004)	0.673 (0.001)	2.449 (0.041)	3.13 (0.17)	0
BAOAB	1.22 (0.005)	0.673 (0.002)	2.452 (0.045)	3.14 (0.22)	0
FLA	0.194 (0.032)	0.060 (0.013)	1.124 (0.037)	0.011 (0.004)	6.3×10^{-5}
PT	0.364 (0.028)	0.119 (0.015)	1.063 (0.056)	0.039 (0.008)	0
CP	0.410 (0.021)	0.077 (0.016)	2.085 (0.027)	0.017 (0.006)	0.209
LSC-CP	0.124 (0.031)	0.028 (0.013)	1.321 (0.126)	0.0023 (0.0018)	0.0036
LSC-CP-MA	0.100 (0.028)	0.022 (0.011)	1.294 (0.088)	0.0015 (0.0013)	0.0029

Table 2: Terminal metrics at $t = T$, mean with SD shown in parentheses where available over seed blocks; entries without parentheses are rounded means. W_2: sliced (exact in 1D) Wasserstein-2 against the reference in metric space; $\mathrm{TV}_{\mathrm{basin}}$: total variation of basin occupancy; $\mathrm{RMSE}_{F/(k_{\mathrm{B}}T)}$: dimensionless additive-aligned reduced-free-energy-surface RMSE (numerically an energy error in $k_{\mathrm{B}}T$) on the two-dimensional collective variables (one-dimensional for the double well); D_{KL}: target-to-empirical basin KL with Jeffreys pseudocount; FES outside: empirical mass outside the frozen reference FES grid. Metric definitions in Appendix C.

the reference free-energy grid is 31.6%/17.2%/20.9% on those systems, versus $\leq 0.62\%$ for LSC-CP. Raw CP also contacts the finite box on 5.1%, 6.7%, and 0.4% of applied jumps on these systems (Appendix E); this is a numerical-box confound, especially for MoG40 and MB, rather than an intrinsic component of the stationary defect. In one ϕ^4 sensitivity comparison, widening the box from $[-2,2]^{24}$ to $[-5,5]^{24}$ changes raw-CP W_2 from 0.28 to 0.41, whereas LSC-CP-MA changes from 0.099 to 0.100. This single check supports robustness to that particular box change but does not establish general box-size independence.

(iv) Random-bank and deterministic estimators are qualitatively consistent. Where both are run (double well and MoG40), the paired single-atom estimator preserves the qualitative method ordering and gives comparable terminal errors on several metrics. It is not numerically equivalent on every metric: for example, its MoG40 basin TV is 0.111 versus 0.053, its KL is 0.0448 versus 0.0089, and its outside-grid mass is about four times larger. In MB and ϕ^4, deterministic and paired multi-atom LSC estimates have similar terminal W_2, TV, KL, and FES values, but both implementations retain energy discrepancies large relative to their chain MCSEs. These comparisons provide estimator-sensitivity checks, not proofs of numerical equivalence.

5 Post-settling time-series diagnostics

Terminal ensemble metrics quantify the law at time T; they do not measure the precision of time averages. For that purpose, small independent chains (4 seeds × 8 chains, 1000 uniformly spaced draws over one additional horizon) are initialized from the system reference, evolved for one full settling horizon, and then analyzed for basin indicators, energy, and collective variables. None is claimed to be an exact stationary draw of the discrete kernel, so the values below are post-settling diagnostics of the numerical chains (Appendix C). A constant basin indicator contributes zero ESS by construction. Raw CP and FLA are excluded because their invariant laws are not π_β.

	worst-basin ESS (of 32,000)				energy ESS / signed energy bias			
Method	DW	MoG40	MB	ϕ^4	DW	MoG40	MB	ϕ^4
ULA	0	0	0	0	14,815 / +0.0003	707 / +0.001	22,223 / +0.007	17,212 / +0.133
MALA	0	0	0	0	14,869 / +0.0005	710 / −0.004	21,994 / −0.001	15,250 / −0.010
BAOAB	0	0	0	0	3,658 / −0.0002	534 / −0.001	5,622 / +0.004	2,734 / +0.028
PT	1,211	572	2,336	2,567	20,494 / −0.001	1,340 / +0.0006	25,202 / −0.0004	24,383 / −0.002
LSC-CP	1,648	271	2,066	1,222	31,891 / +0.194	24,051 / +0.166	27,993 / +0.040	30,475 / +0.255
LSC-CP-RA	1,809	859	—	—	31,224 / +0.247	8,119 / +0.717	—	—
LSC-CP-MA	—	—	1,952	1,150	—	—	26,720 / +0.039	30,972 / +0.251

Table 3: Post-settling trace diagnostics aggregated over all seeds and chains. Worst-basin ESS is the minimum, over basins, of the summed per-chain effective sample size of that basin's indicator (Geyer IAT; constant chains contribute zero). Energy bias is the signed difference between the trace mean of V and the reference target, in energy units; it is interpreted with its observable-specific Monte Carlo standard error. The deterministic/numerical LSC-CP row and paired LSC-CP-MA row are shown separately for MB and ϕ^4. Split-$\widehat{R}$ and MCSE values for every observable are reported in the supporting data.

The deterministic/numerical LSC energy MCSEs are 0.0194 (double well), 0.00593 (MoG40), 0.00125 (MB), and 0.0148 (ϕ^4); the paired multi-atom values are 0.00143 (MB) and 0.0141 (ϕ^4), and the double-well and MoG40 single-atom values are 0.0273 and 0.0235. Energy split-$\widehat{R}$ is below 1.11 for the tabulated methods, but this does not extend to every basin or collective-variable trace. For example, maximum values are 1.46 for MoG40 LSC-CP, 1.33 for MoG40 LSC-CP-RA, 1.28 for

MoG40 PT, and 1.13 for ϕ^4 PT. We therefore use these quantities as finite-horizon diagnostics and interpret them jointly with population errors.

Three conclusions follow from Table 3.

(i) Between-basin ESS separates the methods qualitatively. The three local integrators record *exactly zero* worst-basin ESS on every system because at least one basin indicator is constant or unvisited in the recorded traces. This is not equivalent to zero transitions: on MoG40, for example, ULA, MALA, and BAOAB record 222, 230, and 60 label changes, and the corresponding MB counts are 209, 194, and 43. Their large energy ESS values illustrate why energy decorrelation alone is insufficient evidence of global mixing. LSC-CP sustains worst-basin ESS between 271 and 2,066 for the deterministic/numerical implementation and between 1,150 and 1,952 for the higher-dimensional multi-atom implementation—the same order as fully costed parallel tempering—using local gradients and the jump-bank score.

(ii) Mixing and accuracy must be read together. On MoG40, the worst-basin ESS is 572 for PT, 271 for numerical LSC-CP, and 859 for the single-atom estimator; on ϕ^4 it is 2,567 for PT, 1,222 for deterministic LSC-CP, and 1,150 for the multi-atom estimator. The LSC implementations nevertheless have smaller terminal basin KL at the reported horizon (Table 2). The ϕ^4 population difference is reference-sensitive as discussed above. Neither diagnostic alone ranks the samplers; population error, observable bias, mixing, and computational cost must be reported together.

(iii) The LSC chains carry a material energy discrepancy. The post-settling energy mean of the deterministic and multi-atom LSC chains exceeds the frozen target estimate by +0.039 to +0.255 energy units; these differences are 17–32 times the reported chain-mean MCSE. The MCSE does not propagate uncertainty in the numerical reference energy, so this comparison is conditional on the frozen reference estimate and is not a complete significance calculation. The single-atom estimator has discrepancies +0.247 (double well) and +0.717 (MoG40), so random-bank pairing or randomization does not by itself control invariant-measure bias. Local methods also show nonzero finite-run differences from the frozen reference estimate: on ϕ^4, these are +0.1334 for ULA, −0.0100 for MALA, and +0.0276 for BAOAB. In particular, the MALA value should not be attributed to transition-kernel bias without propagating chain, reference, and finite-box uncertainty. The LSC discrepancy cannot be attributed merely to high-energy post-jump chord configurations, because those transients also occur in the exact continuous-time process whose drift–jump balance preserves π_β. Instead, it indicates observable-dependent invariant-measure error in the finite-step kernel, potentially from Euler splitting, drift taming, finite quadrature, exponent capping, or box projection. These devices are useful for robust computation, but systematic refinement is needed to separate their contributions to the measured bias.

Cost-normalized throughput. Using the wall-clock ledger in the current stationarity artifacts, MB energy ESS/s is 14.10 for PT, 5.95 for deterministic LSC-CP, and 6.97 for LSC-CP-MA; the corresponding worst-basin values are 1.307, 0.439, and 0.509. On ϕ^4, energy ESS/s is 10.67/5.14/6.28 and worst-basin ESS/s is 1.124/0.206/0.233 in the same method order. Per particle step, deterministic LSC uses $Aq_\rho q_\theta = 128/1024$ score-chord energy differences on MB/ϕ^4, whereas the paired multi-atom estimator uses $Aq_\theta = 32/256$. Per-gradient, per-potential, and per-score-quadrature ESS are stored separately; the wall-clock values are single-run, hardware-dependent measurements rather than uncertainty-qualified performance constants (Appendix F).

6 Scope and limitations

Three boundaries of the present evidence are stated explicitly. First, all results are equilibrium-sampling diagnostics of algorithmic chains; passage or round-trip statistics, where recorded, are Kaplan–Meier restricted means of algorithmic mixing times and carry no molecular-kinetics meaning. Second, the finite-step LSC chains are not exactly π_β-invariant, and the measured consequences—the energy bias of Section 5 and the clipping/boundary diagnostics of Appendix E—are part of the reported results. Third, the present comparison does not include a matched-length *random-geometry* jump control (same atom count, length multiset, weights, rate, and cost; randomized directions) that would isolate the contribution of the designed bank geometry from the generic value of long jumps; raw CP controls for the score correction, not for geometry. Box sensitivity has been checked only for one ϕ^4 change ($L = 2$ to $L = 5$); broader checks at $L = 5, 6, 7$ are needed. Finally, the SNIS and long-PT ϕ^4 references differ by basin TV 0.042, so population-error distinctions below that scale are reference sensitive. The legacy sample-based FES plotting helper also computes $-\log p/\beta$ while labelling the color scale in $k_\mathrm{B}T$; those saved FES profile/map color bars are compressed by a factor β and should be regenerated before quantitative use. This plotting defect does not affect the tabulated augmented-grid FES RMSE, which is computed independently from $-\log p$.

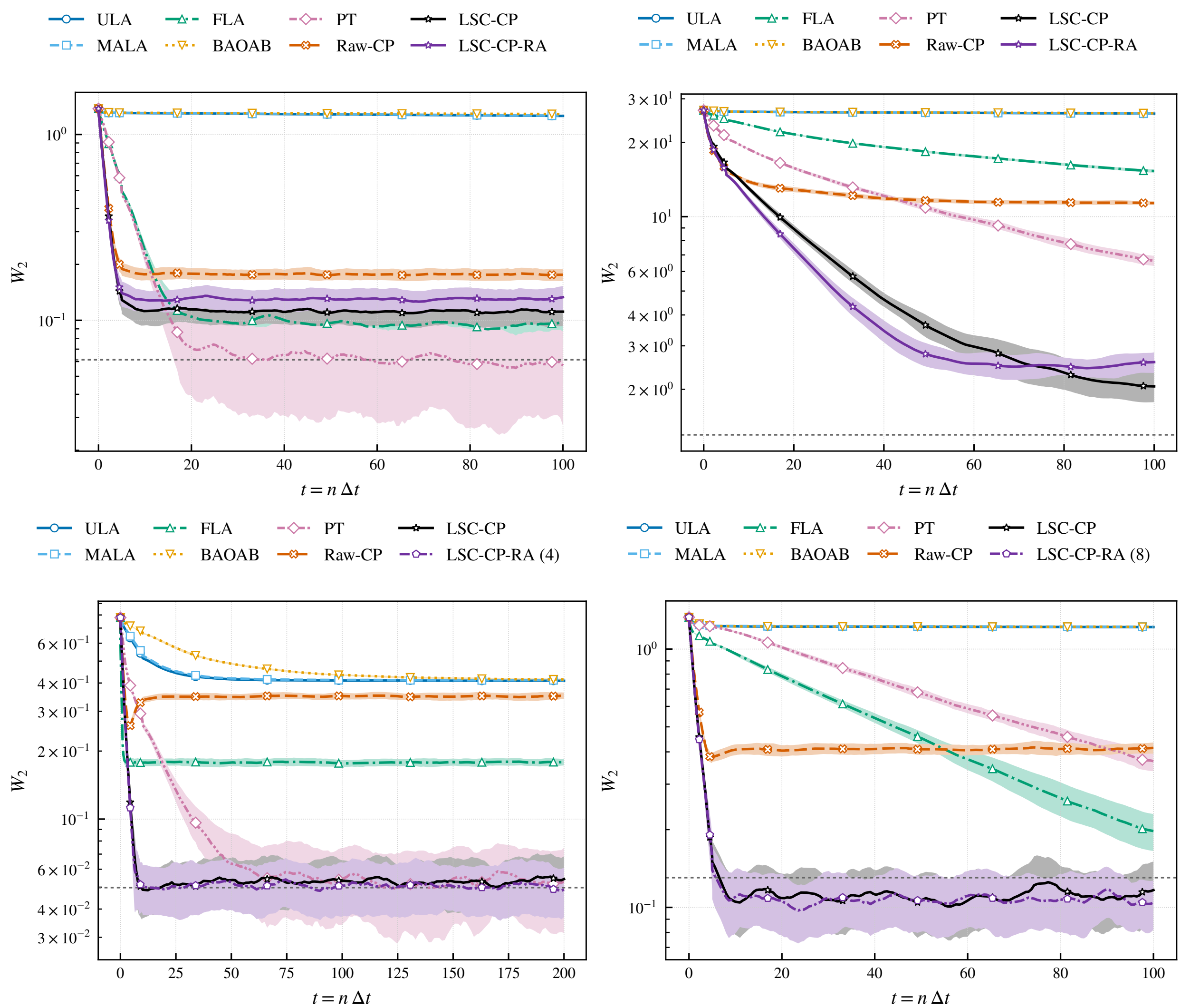

Figure 1: Ensemble relaxation: W_2 to the reference versus simulation time t from the shared localized initial ensembles. Curves and shaded envelopes are centered nine-checkpoint running averages of the seed-block mean and mean $\pm$ one seed-block SD; the dashed horizontal line is the reference mean, not the $\mu_{\text{ref}} + 3s_{\text{ref}}$ descriptive upper band. Top: double well, MoG40; bottom: Müller–Brown 10D, coupled ϕ^4.

A The Lévy score and the paired-bank estimators

For a finite jump measure ν, the jump generator is $(J_\nu f)(x) = \int [f(x+r) - f(x)]\, \nu(\mathrm{d}r)$, and the Lévy score is

$$S_{\nu,\beta}(x) = -\int_{\mathbb{R}^d\setminus\{0\}} r \int_0^1 \frac{\pi_\beta(x-\theta r)}{\pi_\beta(x)}\, \mathrm{d}\theta\, \nu(\mathrm{d}r) = -\int r \int_0^1 e^{\beta[V(x)-V(x-\theta r)]}\, \mathrm{d}\theta\, \nu(\mathrm{d}r). \tag{7}$$

The chord fundamental theorem of calculus gives, for smooth compactly supported f,

$$\int S_{\nu,\beta}(x)\cdot \nabla f(x)\, \pi_\beta(x)\, \mathrm{d}x = -\int (J_\nu f)(y)\, \pi_\beta(y)\, \mathrm{d}y, \tag{8}$$

hence $\int A_{\nu,\beta} f\, \pi_\beta\, \mathrm{d}x = 0$ for the generator (2): an infinitesimal, continuous-time weak stationarity statement.

For a finite bank $\nu_{\mathcal{R}}(\mathrm{d}r) = \lambda \sum_{a=1}^{A} w_a \delta_{R_a}(\mathrm{d}r)$ the score is

$$S_{\mathcal{R},\beta}(x) = -\lambda \sum_{a=1}^{A} w_a R_a I_a(x), \qquad I_a(x) = \int_0^1 e^{\beta[V(x)-V(x-\theta R_a)]}\, \mathrm{d}\theta, \tag{9}$$

with the chord integral evaluated by Gauss–Legendre quadrature. For the deterministic shell calculation, radial quadrature additionally replaces the continuous shell law by a finite quadrature measure; the full-law jump step still samples the continuous shell. This is a separate approximation from time discretization. The reported MoG40 artifact applies Gauss–Legendre quadrature in chord and radius and a periodic trapezoidal rule in angle. With $u_\ell = (\cos(2\pi\ell/m_\phi), \sin(2\pi\ell/m_\phi))$, normalized radial weights $\widehat{w}^\rho_j$, and normalized chord weights $\widehat{w}^\theta_p$, its deployed score is

$$S_Q(x) = -\frac{\lambda}{m_\phi} \sum_{\ell=0}^{m_\phi-1} \sum_{j=1}^{q_\rho} \widehat{w}^\rho_j\, \rho_j u_\ell \sum_{p=1}^{q_\theta} \widehat{w}^\theta_p e^{\beta[V(x)-V(x-\theta_p\rho_j u_\ell)]}. \tag{10}$$

The saved artifacts use $(q_\theta, q_\rho, m_\phi) = (16, 4, 16)$; the current builder default is $(16, 4, 64)$ for a future rerun. A closed-form chord–radius `MoG40Score` remains in the source tree as a validation comparator, but it is not the deployed production estimator.

The score is accumulated in log space. For each particle, the largest log magnitude is extracted over the positive quadrature/atom contributions *before* their signed vector sum is formed. Rescaling by this common maximum preserves cancellation among opposite vectors. The implementation then replaces the common magnitude $\exp(M)$ by $\exp[\min(M, M_{\max})]$, with $M_{\max} = 600$. When active, this cap preserves the relative atom contributions and the score direction but changes its overall magnitude. It is therefore a numerical-stability device, not an algebraically exact score evaluation. Its activation frequency and the maximum pre-cap exponent are recorded in Appendix E.

A.1 Paired random banks

When the intended law is a mixture of conditional atom laws, $\nu(\mathrm{d}r) = \lambda \sum_a w_a q_a(\mathrm{d}r)$, the paired multi-atom estimator draws, at the start of each step and independently of the state,

$$R_{n,a} \sim q_a, \qquad \widehat{\nu}_n(\mathrm{d}r) = \lambda \sum_{a=1}^{A} w_a \delta_{R_{n,a}}(\mathrm{d}r), \qquad \mathbb{E}[\widehat{\nu}_n] = \nu, \tag{11}$$

evaluates a finite-q_θ approximation to (9) for the *realized* bank, and draws the jump counts conditionally on that same bank,

$$N_{n,a} \sim \text{Poisson}(\lambda w_a \Delta t), \qquad \Delta J_n = \sum_{a=1}^{A} N_{n,a} R_{n,a}. \tag{12}$$

With the exact chord integral, each frozen bank would satisfy the weak drift–jump identity (8). The finite chord rule makes this identity approximate, but pairing still removes the additional mismatch that would arise if the score and jumps used independent banks. Averaging over the atom draw makes the estimator unbiased for the *quadrature-defined* score, not for the exact continuous-score integral. The single-atom variant LSC-CP-RA is the $A = 1$ case with the atom family resampled each step.

The implemented step, with integration box $\mathcal{B}$, is

$$\begin{aligned} X_{n+1} = \operatorname{clip}_{\mathcal{B}}\Big[X_n + \Delta t \, \frac{b_n}{1 + \Delta t \|b_n\|/c} &+ \sqrt{2\Delta t/\beta}\, \xi_n \\ &+ \sum_a N_{n,a} R_{n,a} \Big], \\ b_n = -\nabla V(X_n) + S_{\mathcal{R}_n,\beta}(X_n). \end{aligned} \tag{13}$$

The deployed taming scales are $c = 1$ in E1–E2 and $c = 2h$ in E3–E4, where h is the shell half-width. Euler splitting, finite quadrature, exponent capping, drift taming, and box projection all lie outside the exact generator identity. The weak-identity residual tests the uncapped quadrature-level drift–jump cancellation only; it does not test these finite-step modifications. The atomwise counts in (12) are untruncated. The separate full-law compound-Poisson implementation uses a safety cap $K_{\max} = 8$ on applied multiplicity while recording the uncapped Poisson draw; no cap hit occurred in the reported trajectories. Multiple events for one paired atom within a step reuse its stored displacement, an $O(\Delta t^2)$ difference from independently resampling every full-law event.

B System specifications

Double well. $V(x) = (x^2 - 1)^2$, $\beta = 8$; bank: atoms ± 2, weights 1/2, radial half-width 0.2; box $[-5.2, 5.2]$ (widened so that raw CP's excursions do not pile mass at the edge of the density grid).

MoG40. Means μ_k drawn once (fixed seed) in $[-40, 40]^2$; $\pi_\beta = \frac{1}{40}\sum_k N(\mu_k, I_2)$; annulus law with radius uniform on $[10, 14]$, direction uniform; box $[-65, 65]^2$. The artifact/provenance qualification in Section 2 applies.

Depth-balanced Müller–Brown. Latent potential

$$V_{\text{MB3}}(z) = \sum_{k=1}^{4} D_k \exp\{a_k(z_1 - x_k)^2 + b_k(z_1 - x_k)(z_2 - y_k) + c_k(z_2 - y_k)^2\}, \tag{14}$$

k	D_k	a_k	b_k	c_k	x_k	y_k
1	−1.6607	−1	0	−10	1	0
2	−1.0000	−1	0	−10	0	0.5
3	−1.0218	−6.5	11	−6.5	−0.5	1.5
4	0.1500	0.7	0.6	0.7	−1	1

(depths retuned so all three minima sit at $V \simeq -0.7957$). With $x = zB^{\mathsf{T}}$, the full ten-dimensional potential is

$$V_3(x) = V_{\mathrm{MB3}}(z_1, z_2) + \frac{1}{2(0.4)^2} \sum_{j=3}^{10} z_j^2, \qquad z = xB^{-\mathsf{T}}. \tag{15}$$

Here $B = Q\,\mathrm{diag}(0.75, \ldots, 1.45)$, with ten equally spaced diagonal entries and Q obtained by QR factorization of the `default_rng(12345)` Gaussian matrix. The inverse temperature is $\beta = 24$. Bank: four directed relay atoms $A \to B$, $B \to A$, $B \to C$, $C \to B$, weights $1/4$, radial half-width $0.1 \min_a \|r_a\|$; drift cap twice that half-width.

Coupled ϕ^4. Site potential

$$W(u, v) = (u^2 - 1)^2 + (v^2 - 1)^2 - 0.0125\,uv + 0.0075\,u + 0.015\,v, \tag{16}$$

$\kappa = 2.5$, $\delta = 1/12$, twelve periodic sites with $q_{13} = q_1$, $\beta = 8$. Bank: eight coherent edge atoms $\mathbf{1}_{12} \otimes (v_j - v_i)$ along the phase square, weights $1/8$, radial half-width $0.1 \min_a \|r_a\|$, drift cap twice that half-width. The integration box half-width $L = 5$ is *derived*: a simultaneous Laplace-component envelope with union-bound tail budget $\alpha = 10^{-8}$ gives $B_\beta = 1.910$, the hottest-PT envelope $B_{\beta_{\min}=1} = 3.569$, and the maximum componentwise one-jump reach $R_\infty = 2.200$; $L = \lceil \max\{B_\beta + R_\infty, B_{\beta_{\min}}\} \rceil = 5$. The basin map for $\bar{q}$ is constructed by gradient flow on $[-4, 4]^2$—the *jump-reachable* order-parameter set (double-jump reach $|\bar{q}| \leq 3.3$), not merely the phase minima (Appendix E).

Parallel-tempering ladders (auto-tuned, fully costed). Double well: $\beta \in \{8, 1\}$ ($K = 2$, swap acceptance 0.41); MoG40: $\{8, 0.447, 0.025\}$ ($K = 3$, 0.26); MB: $\{24, 13.5, 7.6, 4.3\}$ ($K = 4$, 0.37); ϕ^4: six geometric rungs from 8 to 1 ($K = 6$, 0.31).

Baseline implementation details. ULA, MALA, FLA, and BAOAB use the same finite state box as the corresponding experiment and tame their local drift or force with $c = 1$. MALA uses $h = 2\Delta t/\beta$ and rejects, rather than clips, proposals outside the box. BAOAB uses unit mass and friction $\gamma = 1$. FLA uses independent coordinatewise symmetric α-stable increments with $\alpha = 1.7$. Parallel tempering performs MALA within each replica and alternates adjacent swap pairs every ten steps; reported distributional metrics use the cold replica, while all replica evaluations and wall time are charged.

C Metric definitions

Evaluation coordinates and Wasserstein distance. Let $\xi(x)$ denote the coordinate in which a distributional metric is evaluated. We use $\xi(x) = x$ for E1 and E2, the two active coordinates $\xi(x) = (z_1, z_2)$ for E3, and $\xi(q) = \bar{q} = 12^{-1} \sum_{i=1}^{12} q_i$ for E4. For equal-size empirical samples in one dimension (E1), the exact empirical Wasserstein distance is

$$\widehat{W}_2 = \left[\frac{1}{N} \sum_{i=1}^{N} \big(X_{(i)} - Y_{(i)} \big)^2 \right]^{1/2}. \tag{17}$$

For E2–E4, the table entry labelled W_2 is the sliced distance

$$\widehat{\mathrm{SW}}_2 = \left[\frac{1}{L} \sum_{\ell=1}^{L} W_2^2 \Big((\theta_\ell^\top)_\# \widehat{\mu}_N, (\theta_\ell^\top)_\# \widehat{\nu}_N \Big) \right]^{1/2}, \qquad L = 200. \tag{18}$$

The unit vectors θ_ℓ are drawn once with seed 777 and reused across methods and times. E3 additionally records a ten-dimensional sliced W_2 with a separate frozen projection set (seed 778); it is not the W_2 shown in Table 2. A 500-point Hungarian matching is only a terminal two-dimensional spot check.

Maximum mean discrepancy. The reported MMD is the square root of the biased Gaussian-kernel V-statistic,

$$\widehat{\mathrm{MMD}} = \left[\frac{1}{N^2}\sum_{i,j} k(X_i, X_j) + \frac{1}{N^2}\sum_{i,j} k(Y_i, Y_j) - \frac{2}{N^2}\sum_{i,j} k(X_i, Y_j)\right]^{1/2}. \tag{19}$$

$$k(x, y) = e^{-\|x-y\|^2/(2\sigma^2)}. \tag{20}$$

The diagonal terms are retained. The bandwidth σ is the median pairwise distance among at most 2,048 frozen reference points (seed 99) and is not re-estimated along the trajectory.

Secondary density, energy, and Stein diagnostics. For E1, the density TV uses 350 fixed bins on $[-5.2, 5.2]$:

$$\mathrm{TV}_{\mathrm{density}} = \frac{1}{2}\sum_{b=1}^{350} |\widehat{p}_b - p_b^\star|. \tag{21}$$

This differs from the two-well occupancy TV. The energy-overlap deficit uses 40 frozen energy bins,

$$e_{\mathrm{overlap}} = 1 - \sum_b \min\{\widehat{h}_b, h_b^\star\}, \tag{22}$$

with empirical energy outside the reference grid retained as nonoverlap. The archived kernel Stein discrepancy is the square root of the biased V-statistic

$$\widehat{\mathrm{KSD}} = \left[\frac{1}{m^2}\sum_{i,j=1}^{m} (\mathcal{T}_\pi^x \mathcal{T}_\pi^y k)(X_i, X_j)\right]^{1/2}, \qquad k(x, y) = (1 + \|x - y\|^2)^{-1/2}, \tag{23}$$

where the score is $\nabla \log \pi_\beta = -\beta \nabla V$, $m \leq 512$, and a fixed seed 17 chooses the subsample. KSD can be small for a chain trapped in one mode and is therefore secondary to the basin-aware metrics. Additional first-CV CDF/KDE discrepancies are archived for collaborator parity but do not enter the principal claims.

Basin occupancy, TV, and KL. For basin partition $\{\Omega_k\}_{k=1}^K$ with target masses $p^\star$, $\widehat{p}_k = \frac{1}{N}\sum_i \mathbf{1}\{X_i \in \Omega_k\}$, $\mathrm{TV}_{\mathrm{basin}} = \frac{1}{2}\sum_k |\widehat{p}_k - p_k^\star|$, and

$$D_{\mathrm{KL}}(p^\star \| \widetilde{p}) = \sum_{k: p_k^\star > 0} p_k^\star \log \frac{p_k^\star}{\widetilde{p}_k}, \qquad \widetilde{p}_k = \frac{\widehat{p}_k + \alpha/N}{1 + K\alpha/N}, \quad \alpha = \tfrac{1}{2}. \tag{24}$$

The target-to-empirical orientation penalizes a sampler that misses a target basin; the Jeffreys-scale pseudocount keeps the statistic finite. Basin TV is the total variation of this finite partition and therefore only a lower bound on full-distribution TV. In E2, $p_k^\star = 1/40$ is the operational nearest-center target used by the implementation; it is not asserted to equal the exact Gaussian-mixture probability of every finite Voronoi cell.

The actual partitions are system specific. E1 uses the sign of x; E2 uses the nearest frozen center. E3 constructs a 600^2 gradient-flow label map on $[-2, 1.9] \times [-1.3, 2.6]$ in (z_1, z_2), with flow step 1.5×10^{-4} for 40,000 steps, and integrates target basin masses on a 1200^2 grid. E4 constructs an 800^2 gradient-flow label map of the site potential W on $[-4, 4]^2$ in the $\bar{q}$ plane, using the same flow step and count; its target masses are the direct-SNIS weighted category probabilities. The sign-phase grouping used only to condition the E4 visualization KDE is not this gradient-flow basin partition.

Explored-mode coverage. For the uniform target masses in E1 and E2,

$$\mathrm{EMC} = \frac{\exp[H(\widehat{p})]}{K}, \qquad H(\widehat{p}) = -\sum_k \widehat{p}_k \log \widehat{p}_k. \tag{25}$$

For the nonuniform E3 and E4 target masses, the implementation instead uses

$$\mathrm{EMC} = 1 - \mathrm{JS}_2(\widehat{p}, p^\star), \tag{26}$$

where JS_2 is the base-two Jensen–Shannon divergence. Both definitions have optimum 1, but their numerical values are not comparable across the two cases.

Augmented-grid free-energy RMSE. The current implementation uses a discrete augmented grid, not the earlier $5/N$ reference-weighted definition. Let J be the number of rectangular interior cells and let cell $J+1$ collect all mass outside the frozen grid. With $\alpha = 1/2$, the smoothed empirical and reference probabilities are

$$\widehat{p}_i = \frac{n_i + \alpha}{N + (J+1)\alpha}, \qquad p_i^{\mathrm{ref}} = \frac{n_i^{\mathrm{ref}} + \alpha}{M + (J+1)\alpha}. \tag{27}$$

For E4, $n_i^{\mathrm{ref}} = M \sum_{r:\xi(Q_r)\in C_i} w_r$ is the direct-SNIS effective count; all importance weights are first rescaled to sum to M. The retained set and reduced free-energy difference are

$$\mathcal{I} = \{i : p_i^{\mathrm{ref}} \geq 1/N\} \cup \{J+1\}, \qquad \Delta_i = -\log \widehat{p}_i + \log p_i^{\mathrm{ref}}. \tag{28}$$

The off-grid cell is retained even though its only reference mass may arise from the pseudocount. Uniform cell weights are then used:

$$\mathrm{RMSE}_{F/k_{\mathrm{B}}T} = \left[\frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}}(\Delta_i - c^\star)^2\right]^{1/2}, \qquad c^\star = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}}\Delta_i. \tag{29}$$

This is a dimensionless reduced-free-energy error, numerically equal to an energy error measured in units of $k_{\mathrm{B}}T$. For equal-volume interior cells, $-\log p_i$ and $-\log(p_i/|C_i|)$ differ only by the fitted additive constant; the off-grid cell is a diagnostic discrete state rather than a physical volume element. The raw, unsmoothed empirical off-grid probability is additionally reported as FES outside.

The grids contain 40 bins in E1 and 15^2, 14^2, and 10^2 cells in E2–E4. Each coordinate range is the reference minimum–maximum interval expanded by 5%. E1–E3 use 50,000 reference points for the FES histogram; E4 uses the 200,000 weighted production SNIS proposals. Because the same half-count is combined with different empirical and reference sample totals, and because the off-grid cell is always retained, the metric includes a sample-size-dependent smoothing contribution even when raw outside mass is zero. No finite-sample reference band is currently available for this FES metric.

Energy errors and signed time-series bias. The terminal ensemble reports

$$e_{\mu_V} = \left| \frac{1}{N} \sum_i V(X_i) - \mu_V^\star \right|, \qquad e_{\sigma_V^2} = \left| \frac{1}{N-1} \sum_i (V(X_i) - \overline{V})^2 - (\sigma_V^2)^\star \right|. \tag{30}$$

These are absolute errors. In contrast, a post-settling observable row uses the signed bias $\overline{O} - O^\star$. E4 reference moments are direct-SNIS weighted moments; E1–E3 use the frozen numerical reference sample.

IAT, ESS, MCSE, $\widehat{R}$. Per chain, the integrated autocorrelation time uses Geyer's initial-positive/initial-monotone sequence. Writing $\Gamma_k = \widehat{\rho}_{2k} + \widehat{\rho}_{2k+1}$, pair sums are truncated before the first nonpositive value and replaced by their cumulative minimum $\widetilde{\Gamma}_k$:

$$\widehat{\tau_{\text{int}}} = \max\left\{1, -1 + 2\sum_k \widetilde{\Gamma}_k\right\}. \tag{31}$$

A constant chain is assigned $\tau_{\text{int}} = \infty$ and zero ESS. Independent chains contribute

$$\text{ESS}_{\text{total}} = \sum_m \frac{n_m}{\widehat{\tau_{\text{int}}}_m}, \qquad \text{MCSE} = \frac{s_{\text{pooled}}}{\sqrt{\text{ESS}_{\text{total}}}}. \tag{32}$$

This MCSE quantifies uncertainty of the correlated chain mean only; it does not include uncertainty in the separately estimated reference target $O^\star$. Thus total draws divided by an arithmetic mean IAT is not used. Split-$\widehat{R}$ is rank normalized with average ranks for ties and Blom transform $(r - 3/8)/(S + 1/4)$; the reported value is the maximum of the bulk and folded statistics. Globally constant traces give an undefined $\widehat{R}$. Neither a large ESS nor $\widehat{R}$ close to one proves that the numerical kernel targets π_β.

Finite-sample reference levels and threshold time. Twenty replicates estimate the mean μ_{ref} and sample standard deviation s_{ref} of reference–reference W_2/MMD (and E3 ten-dimensional W_2), or of a fresh reference sample against the target for basin TV, EMC, and E1 density TV. For lower-is-better errors the descriptive band is $\mu_{\text{ref}} + 3s_{\text{ref}}$; it is not a mathematical lower bound. The saved `time_to_threshold_TV` is separately defined as the first checkpoint at which the seed-mean TV is no larger than $2\mu_{\text{ref}}$ for five consecutive checkpoints. It does not use the $\mu + 3s$ band and should not be called a convergence or mixing time.

Cost accounting. The aggregate function-evaluation ledger is

$$\text{NFE} = n_V + n_{\nabla V} + n_{\Delta V}, \tag{33}$$

where $n_{\Delta V}$ counts score-chord potential differences. These three evaluations have different arithmetic and memory costs, so the aggregate is only a bookkeeping convention. PT includes all replicas; reference generation, diagnostics, plotting, and I/O are excluded. Main-ensemble seeds are batched, so their wall time and NFE are batch totals repeated in seed rows, and a zero reported wall-time standard deviation is not timing uncertainty. Stationarity ESS/s uses separately synchronized sampler-only timings.

Passage statistics. Where passage times are recorded they are right-censored at the common horizon, restricted to chains starting in the declared home core, and summarized by the Kaplan–Meier restricted mean; the model-dependent censored-exponential estimate, when saved, is labelled as such. Both are algorithmic-mixing diagnostics only.

D References and the ϕ^4 importance reference

E1–E3. The double-well reference is a dense-grid inverse CDF on the declared box (2×10^5 grid points), and MoG40 permits exact mixture draws. In E3, writing $z = B^{-1}x$, the target factorizes as

$$\pi_z(z) \propto e^{-24V_{\mathrm{MB3}}(z_1,z_2)} \prod_{j=3}^{10} \mathcal{N}\left(z_j; 0, \frac{\sigma_{\mathrm{aux}}^2}{24}\right), \qquad \sigma_{\mathrm{aux}} = 0.4. \tag{34}$$

The reference therefore uses a 2400^2 numerical grid for the active plane and exact Gaussian auxiliary draws. The E3 density panel evaluates and normalizes $e^{-24V_{\mathrm{MB3}}(z_1,z_2)}$ directly on the display grid; it is not a histogram.

E4 production importance reference. The exact marginal of the spatial mean is well defined by

$$p_{\bar{q}}(s) = Z_\beta^{-1} \int \delta^{(2)}\left(s - \frac{1}{12}\sum_{i=1}^{12} q_i\right) e^{-\beta V(q)} \prod_{i=1}^{12} \mathrm{d}q_i, \tag{35}$$

but it has no simple closed form for the coupled quartic field. The numerical reference uses the four-phase Laplace-mixture proposal

$$q_{\mathrm{prop}}(x) = \sum_{k=1}^{4} a_k \mathcal{N}\left(x; m_k, (\beta H_k)^{-1}\right), \qquad a_k \propto e^{-\beta V(m_k)} (\det H_k)^{-1/2}, \tag{36}$$

where $H_k = \nabla^2 V(m_k)$. For proposals $Q_r \sim q_{\mathrm{prop}}$,

$$\widetilde{w}_r = \frac{\exp[-\beta V(Q_r) - \log q_{\mathrm{prop}}(Q_r)]}{\sum_s \exp[-\beta V(Q_s) - \log q_{\mathrm{prop}}(Q_s)]}. \tag{37}$$

The production GPU reference has 2×10^5 proposals, weight ESS 41,182 (20.6%), and maximum normalized weight 3.6×10^{-3}. Basin masses, energy moments and the energy-overlap histogram, CV means, and the FES histogram use these weights directly. The current W_2 and MMD routines accept unweighted point clouds, so those two diagnostics use finite sampling-importance-resampling with oversampling factor 16 and are therefore approximate; this is an implementation choice, not a mathematical restriction on weighted empirical measures.

E4 visualization-only marginal density. The reference panel uses the same model, proposal, proposal count, and nominal seed in an independent CPU reference integration, not the archived CUDA draws. Because CPU and CUDA random streams are backend-specific, its proposal cloud is not bitwise identical: the figure cache has ESS 107,246, whereas the metric reference retains the production ESS 41,182. Let $g(r)$ be the sign phase of $s_r = \bar{q}(Q_r)$, let

$$N_{\mathrm{eff},g} = \left[\sum_{r:g(r)=g} \left(\frac{\widetilde{w}_r}{\sum_{s:g(s)=g} \widetilde{w}_s}\right)^2\right]^{-1}, \tag{38}$$

and let $\widehat{\Sigma}_g$ be the corresponding weighted covariance. The display estimator is

$$\widehat{p}_{\bar{q}}(s) = \sum_{g=1}^{4} \sum_{r:g(r)=g} \widetilde{w}_r \, \varphi_2\left(s; s_r, N_{\mathrm{eff},g}^{-1/3} \widehat{\Sigma}_g\right). \tag{39}$$

On the display region $[-5,5]^2$, the implementation imposes

$$\lambda_{\min}(\widehat{H}_g) \geq [0.65 \min(\Delta x, \Delta y)]^2 \tag{40}$$

on every bandwidth matrix and evaluates the continuous KDE above by weighted binning and FFT convolution on an 801^2 grid. It then truncates negative roundoff artifacts and renormalizes the density. Phase conditioning prevents a global bandwidth from creating artificial bridges between modes; it is a visualization device and does not replace the production basin map or weighted FES histogram.

E Numerical convergence and stability diagnostics

Diagnostic	DW	MoG40	MB	ϕ^4
nonfinite states	0	0	0	0
score-cap fraction, reported LSC	5.02×10^{-4}	0	2.85×10^{-7}	2.09×10^{-4}
state-box fraction, maximum over methods	8.67×10^{-5}	7.92×10^{-4}	3.304×10^{-3}	8.903×10^{-4}
jump-boundary contact, reported LSC	1.69×10^{-4}	2.13×10^{-5}	2.18×10^{-3}	1.60×10^{-4}
jump-boundary contact, raw CP	3.03×10^{-4}	5.117×10^{-2}	6.733×10^{-2}	4.130×10^{-3}
maximum basin-map outside mass, targeting rows	0	0	0	10^{-3}
uncapped weak-identity residual	1.5×10^{-7}	2.1×10^{-8}	5.0×10^{-7}	2.0×10^{-7}

Table 4: Numerical-stability measurements for the reported trajectories. Score-cap and state-box entries are fractions of particle proposals; the state-box row is the maximum over all compared methods. Jump-boundary contact is measured per applied jump. The basin-map row is the maximum over all targeting-method rows and recorded checkpoints.

The score-cap activation is small but not identically zero for the double well, MB, or ϕ^4. On the deliberately extended weak-identity grids, the fractions with pre-cap exponent above $M_{\max}$ are 0.402 for the double well and 0.374 for MB; these grid values are distinct from the trajectory activation frequencies in Table 4. The residual itself is formed from the uncapped log-domain score and tests the generator identity for the quadrature-defined measure and a finite family of observables. It does not test the exponent cap, drift taming, Euler splitting, or box projection, and it is not a certificate of the invariant distribution of the stabilized finite-step chain.

Because compound-Poisson jumps occur independently of the current state, rare events can contact any finite computational box. The contact and projection frequencies in Table 4 quantify this exposure. In particular, the larger raw-CP contact rates on MoG40 and MB make the finite box a material confound in that ablation. For ϕ^4, the basin map was constructed on $[-4,4]^2$ to cover the jump-reachable order-parameter region rather than only the phase minima. Its targeting-method maximum is 10^{-3}: one of $N = 1000$ particles lies outside the map at the relevant checkpoint for both deterministic LSC-CP and LSC-CP-MA, while the terminal seed-block mean returns to zero (raw CP reaches 5×10^{-3}). Even rare cap, projection, or map-domain events matter for interpretation, so low frequencies are evidence of numerical robustness, not exact stationarity.

F Computational details and data availability

The production manifests record float64 arithmetic on one NVIDIA H200 GPU, Python 3.12.13, and PyTorch 2.13.0+cu130. The repository mirror contains the terminal positions, metric time series, terminal summaries, manifests, and per-observable stationarity summaries. It does not currently mirror the raw stationarity trace arrays or the original/resolved run configurations named by the absolute paths in the manifests; consequently, ESS and provenance cannot be recomputed from the clone alone. Gradient, potential, and Lévy-score-quadrature evaluation counts are nevertheless stored separately for each reported observable.

The MB and ϕ^4 ESS/s values in Section 5 use the synchronized sampler-only timings stored in the current stationarity summaries. They are single-run measurements and are not accompanied by timing replicates. Source code, complete immutable run artifacts, resolved configurations, and analysis-ready numerical data should be deposited together in the publication archive.

G Alanine dipeptide under a real force field (E5)

This appendix documents a fifth, *supplementary* experiment. The four systems of Section 2 remain the reported benchmark; E5 is a single demonstration that the construction can be driven by a real molecular force field rather than an analytic model potential, and it is deliberately placed here rather than in the main text. Nothing in Sections 4–5 depends on it. The purpose of the appendix is completeness of the setup: the description below is intended to be sufficient to reproduce the construction. The vocabulary of molecular simulation is defined as it appears; no prior background is assumed.

G.1 The system and the sampling target

The molecule. Alanine dipeptide (Ac-Ala-NHMe) is a 22-atom molecule consisting of a single alanine residue capped at both ends. A *residue* is one repeating unit of a protein chain; alanine dipeptide is the smallest fragment that still contains a complete protein backbone segment, and it is the standard minimal test system of the molecular-simulation literature. We use `openmmtools.testsystems.AlanineDipeptideVacuum` with `constraints=None`, i.e. the *fully flexible* molecule: every chemical bond is an ordinary spring and no degree of freedom is rigidly constrained. The molecule sits in vacuum (no solvent, no periodic box, no electrostatic cutoff). The configuration space is therefore all $3N = 66$ Cartesian atom coordinates and the potential energy is a smooth function on $\mathbb{R}^{66}$.

What a force field is. A *force field* is an explicit, closed-form formula for the potential energy of a molecule as a function of its atom positions, with parameters fitted once and for all to quantum-chemical and experimental data. It replaces an electronic-structure calculation by a sum of simple mechanical terms. In AMBER convention, and in OpenMM units (energy kJ/mol, length nm, charge e, angles rad), the terms present in this system are

$$U_{\text{bond}} = \sum_{\text{bonds}} \tfrac{1}{2}k(r - r_0)^2, \qquad \text{21 terms,} \tag{41}$$

$$U_{\text{angle}} = \sum_{\text{angles}} \tfrac{1}{2}k(\vartheta - \vartheta_0)^2, \qquad \text{36 terms,} \tag{42}$$

$$U_{\text{tors}} = \sum_{\text{torsions}} k\big[1 + \cos(n\varphi - \varphi_{\text{ph}})\big], \qquad \text{52 terms,} \tag{43}$$

$$U_{\text{nb}} = \sum_{i<j \text{ not excepted}} \Big[\frac{k_e\, q_i q_j}{r_{ij}} + 4\epsilon_{ij}\Big(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^{6}}{r_{ij}^{6}}\Big)\Big]. \tag{44}$$

Here r is a bond length (distance between two bonded atoms), ϑ a bond angle (the angle at the central atom of three consecutively bonded atoms), and φ a torsion angle, defined below. The nonbonded term combines Coulomb attraction/repulsion between partial atomic charges q_i with a Lennard-Jones term describing short-range repulsion and dispersion; the mixing rule is Lorentz–Berthelot, $\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j)$, $\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}$. The Coulomb constant $k_e = 138.93545764438\ \mathrm{kJ\,nm\,mol^{-1}e^{-2}}$ is probed from OpenMM at run time rather than hard-coded. Ninety-eight pair *exceptions* override the nonbonded formula for pairs that are already described by bonded terms: pairs separated by one or two bonds are excluded entirely ($q_i q_j = \epsilon_{ij} = 0$) and pairs separated by three bonds use scaled parameters. The total energy is $U = U_{\mathrm{bond}} + U_{\mathrm{angle}} + U_{\mathrm{tors}} + U_{\mathrm{nb}}$. This is the first system in the note whose V is not an analytic model surface.

The target. The temperature is $T = 300\,\mathrm{K}$. Writing k_{B} for Boltzmann's constant, the thermal energy scale is $k_{\mathrm{B}}T = 2.4943$ kJ/mol and $\beta = 1/(k_{\mathrm{B}}T) = 0.40091$ mol/kJ, derived from the molar gas constant rather than assigned. Energies quoted in units of $k_{\mathrm{B}}T$ are *dimensionless* energies; a barrier of $1\,k_{\mathrm{B}}T$ is comparable to the ambient thermal fluctuation, while a barrier of several $k_{\mathrm{B}}T$ is crossed only rarely. The target is the Cartesian Boltzmann law

$$\mu(\,\mathrm{d}x) \propto e^{-\beta U(x)}\,\mathrm{d}x, \qquad x \in \mathbb{R}^{66}, \tag{45}$$

with U the full force field: nothing is coarse-grained and no degree of freedom is integrated out. The suite-wide constant $\beta = 8$ used elsewhere in this note is untouched; E5 threads its own physical β through the run configuration exactly as the Müller–Brown system threads $\beta = 24$.

Torsion angles and the two collective variables. Given four consecutively bonded atoms p_1–p_2–p_3–p_4, the *dihedral* (equivalently *torsion*) angle is the rotation angle about the central p_2–p_3 bond: the angle between the plane through $p_1p_2p_3$ and the plane through $p_2p_3p_4$. Rotating a torsion swings an entire molecular fragment rigidly about a bond axis. The two backbone torsions of alanine dipeptide are

$$\begin{aligned} \phi &= \mathrm{dihedral}(\mathrm{ACE\!:\!C},\ \mathrm{ALA\!:\!N},\ \mathrm{ALA\!:\!CA},\ \mathrm{ALA\!:\!C}), \\ \psi &= \mathrm{dihedral}(\mathrm{ALA\!:\!N},\ \mathrm{ALA\!:\!CA},\ \mathrm{ALA\!:\!C},\ \mathrm{NME\!:\!N}), \end{aligned} \tag{46}$$

i.e. atom index sets $(4, 6, 8, 14)$ and $(6, 8, 14, 16)$. Both were selected by residue and atom *name* and then verified to reproduce, index for index, the sets returned by `mdtraj`'s `compute_phi`/ `compute_psi`, with angle values agreeing to below 10^{-6} rad. Each of ϕ, ψ is 2π-periodic, so (ϕ, ψ) lives on a two-dimensional torus; a plot of the free energy over that torus is called a *Ramachandran plot* and is the standard description of backbone conformation. Bond lengths and bond angles are stiff and vibrate rapidly about fixed values; the backbone torsions are the soft, slow coordinates. They are therefore the natural *collective variables* (low-dimensional functions of the configuration used to describe and to measure the slow motion) for this system, and they play the same role that the latent plane plays for Müller–Brown and the spatial mean $\bar{q}$ plays for ϕ^4.

G.2 Why the system is a meaningful test

A *metastable state* is a region of configuration space in which the dynamics remains trapped for a long time relative to the local relaxation time, because leaving it requires passing a free-energy barrier. The *free energy* of a collective variable value s is $F(s) = -\beta^{-1} \log \int \delta(s(x) - s)\, e^{-\beta U(x)}\,\mathrm{d}x$ up to an additive constant; $F(\phi, \psi)$ is what a Ramachandran plot displays, and a *free-energy surface* (FES) is this function on a grid.

Alanine dipeptide in vacuum has three metastable states, denoted C5/β, C7eq and C7ax. Two of them, C5/β and C7eq, sit at negative ϕ and together carry approximately 99% of the equilibrium probability. The third, C7ax, sits at *positive* ϕ and carries under 1%. Reaching and correctly weighting this island is the slow event of the system: its escape barrier is about $4.8\,k_{\mathrm{B}}T$, i.e. 12.0 kJ/mol. This is the same qualitative challenge as the low-mass phases of the ϕ^4 chain, but here the landscape is not designed—it is whatever the force field produces.

One feature is not obvious and is important for the construction of the jump bank. The cheap route between the negative-ϕ states and the positive-ϕ island does *not* pass through $\phi = 0$. Crossing the $\phi = 0$ dividing line costs 32.5 kJ/mol, whereas the island's own escape barrier is only 12.0 kJ/mol and is reached near $\phi = \pm 180°$, i.e. around the far side of the torus. A jump geometry that measures displacements with the torus-minimal (periodic) convention takes the cheap route automatically; a jump geometry that treats ϕ as a coordinate on the line would not.

G.3 Why a straight-line Cartesian jump must fail

Every estimator in this note evaluates the Lévy score (9) along the straight *chord* $\theta \mapsto x - \theta r$, $\theta \in [0, 1]$. In Cartesian atom coordinates this is fatal for a molecular transition, for a purely geometric reason.

A basin-to-basin transition of alanine dipeptide is, physically, a *rotation* of a molecular fragment by an angle Δ about a backbone bond axis. Consider an atom of the rotating fragment at perpendicular distance ρ from that axis. Under the true motion it travels along a circular arc of radius ρ; its distance to the axis is ρ throughout. A jump $x \mapsto x + r$ that is a straight line in $\mathbb{R}^{66}$ instead follows the *chord* of that arc. Identifying the plane perpendicular to the axis with $\mathbb{C}$, the chord point at fraction θ has distance to the axis

$$\rho\big|(1-\theta) + \theta e^{i\Delta}\big| = \rho\sqrt{(1-\theta)^2 + \theta^2 + 2\theta(1-\theta)\cos\Delta}, \tag{47}$$

which is minimized at $\theta = \frac{1}{2}$ with value $\rho\cos(\Delta/2)$. The chord therefore contracts the atom towards the axis by a radial deficit

$$\rho\big[1 - \cos(\Delta/2)\big] = \tfrac{1}{8}\rho\,\Delta^2 + O(\Delta^4). \tag{48}$$

For a realistic backbone transition, $\Delta \approx 140°$ and $1 - \cos 70° \approx 0.66$: at the midpoint of the chord the atom is pulled 66% of the way to the axis. Bonds and bond angles that cross the rotation boundary are stiff springs, and they must absorb the entire contraction. The chord is consequently driven hundreds of $k_{\mathrm{B}}T$ above both basins, the score integrand $e^{\beta[U(x)-U(x-\theta r)]}$ in (9) saturates the exponent cap $M_{\max} = 600$ of Appendix A, and the score is no longer an evaluation of the intended quantity. This is a structural obstruction, not a tuning difficulty: it is present for *every* conformational jump of *every* flexible molecule expressed in Cartesian coordinates.

Equation (48) also contains the escape route. The deficit is $O(\Delta^2)$ in the *rotation angle*, not in the Cartesian displacement. A coordinate system in which a conformational transition is an affine displacement—so that the chord is the rotation itself, at $\Delta = 0$ deficit—removes the obstruction entirely. That is why E5 is run in internal coordinates.

G.4 Internal (BAT) coordinates

Construction. BAT stands for bond/angle/torsion. Instead of three Cartesian numbers per atom, each atom after the first three is placed relative to three already-placed reference atoms by a triple (b, a, τ): a bond length b to its bond parent, a bond angle a at that parent, and a torsion τ about the parent–grandparent axis. The resulting table is a *Z-matrix*, and the reconstruction

$q \mapsto x$ is performed by the standard NeRF (natural extension reference frame) recursion. Three root atoms fix the six external degrees of freedom (three translations and three rotations) on which U does not depend. The internal dimension is therefore

$$d = 3N - 6 = 60 = \underbrace{21}_{\text{bonds}} + \underbrace{20}_{\text{angles}} + \underbrace{19}_{\text{torsions}} . \tag{49}$$

The root is (ACE : C = 4, ALA : N = 6, ALA : CA = 8), and the first two non-root atoms are forced to be ALA : C = 14 and NME : N = 16 so that their torsions *are* the collective variables: $\phi = q_5$ and $\psi = q_8$ in zero-based slot indexing.

Jacobian. Changing coordinates changes the volume element, and the *Jacobian determinant* $|\det(\partial x_{\text{free}}/\partial q)|$ is the local volume ratio. Placing one atom by (b, a, τ) is exactly a spherical-coordinate parametrization about its reference atom, whose Cartesian element is $b^2 \sin a \, \mathrm{d}b \, \mathrm{d}a \, \mathrm{d}\tau$. Composing over atoms gives

$$\log\bigl|\det(\partial x_{\text{free}}/\partial q)\bigr| = \log r_{23} + \sum_{\text{non-root } i} \bigl[2 \log b_i + \log \sin a_i\bigr], \tag{50}$$

where the frame boundary term $\log r_{23}$ comes from the third root atom, which contributes a two-dimensional polar element (one power of the bond, no sine). Equation (50) was checked against the numerical log-determinant of the full 60×60 Jacobian and agrees to 1.4×10^{-14}. The target in internal coordinates is

$$\mu_q(\,\mathrm{d}q) \propto e^{-\beta U_{\text{eff}}(q)} \, \mathrm{d}q, \qquad U_{\text{eff}}(q) = U(x(q)) - \beta^{-1} \log|\det(\partial x_{\text{free}}/\partial q)|. \tag{51}$$

The pushforward of μ_q under $q \mapsto x$ is exactly the Cartesian Boltzmann law (45), so this is the *same* target and the same measure the reference of Appendix G.5 samples. We emphasize what this is not: it is a change of sampling variables, not a model of curved-space kinetics. The dynamics is identity-metric Langevin on U_{eff}; there is no position-dependent mass metric and no Fixman correction. Because the object of study is the invariant measure, this is correct and is the simplest choice, and it is applied identically to every compared method.

Torsion independence and the Jacobian-free chord. No term of (50) contains a torsion: the azimuth enters the spherical element only through $\mathrm{d}\tau$ and never as a multiplicative weight. The measured maximum torsion derivative of $\log|\det J|$ is exactly 0.0. The consequence is the property that makes the whole experiment possible. If a jump displacement r is nonzero *only* in torsion slots, then every bond and every angle is unchanged along the entire chord $q - \theta r$, so $\log|\det J|$ is constant along it and cancels in the score integrand:

$$U_{\text{eff}}(q - \theta r) - U_{\text{eff}}(q) = U\bigl(x(q - \theta r)\bigr) - U\bigl(x(q)\bigr). \tag{52}$$

The Lévy score therefore needs only *physical energy differences* along the chord—precisely what a black-box force-field evaluator returns—and never the Jacobian, its gradient, or its second derivatives. The measured residual of (52) is 4.6×10^{-13}, i.e. machine zero.

Leader and offset torsions. A naive Z-matrix is not yet adequate, and the failure mode is quantitative. Several atoms can share the same bond parent and the same angle parent, so they rotate about the *same* axis. If each such atom carries an absolute torsion, then moving one of them while holding its siblings fixed does not rotate a fragment—it distorts the local geometry. Moving

ϕ would carry ALA:C and its subtree while leaving HA and CB behind, breaking the tetrahedral centre at CA. The measured curvature of U_{eff} in ϕ under that parametrization is $822\ \text{kJ mol}^{-1}\text{rad}^{-2}$ against roughly 80 for the true soft rotation: the Cartesian chord pathology of Appendix G.3 reappearing inside internal coordinates. The fix is to designate the first atom placed about each axis the *leader*, carrying the proper torsion, and to let its siblings carry the *offset* from the leader, which is stiff and nearly constant. There are 8 leader and 11 offset torsions. The reparametrization is unit triangular in the torsion block, so $\log|\det J|$ is unchanged and (50)–(52) continue to hold verbatim.

Whitening. Bond and angle modes are stiff (force constants from hundreds to hundreds of thousands in the above units), so a single global timestep in raw q would be hostage to the fastest mode. We absorb a *fixed diagonal* whitening into the coordinates, exactly as the Müller–Brown and ϕ^4 systems absorb their fixed embedding: the sampler coordinate is $\tilde{q} = D^{-1}q$ and the sampler potential is $V(\tilde{q}) = U_{\text{eff}}(D\tilde{q})$. Isotropic Langevin in $\tilde{q}$ is preconditioned Langevin in q and preserves the same target, since D is a constant linear change of variables. The entries are

$$D_{ii} = 1 \text{ for the } \phi \text{ and } \psi \text{ slots,} \qquad D_{ii} = \min\Big\{1,\ \sqrt{1/(\beta H_{ii})}\Big\} \text{ otherwise,} \tag{53}$$

with H the diagonal of the U_{eff} Hessian. Keeping $D_{ii} = 1$ on ϕ and ψ means those two coordinates remain affine and unit scale, so a jump atom reads directly as a rotation in radians. Provenance matters here: the supplied conformer is *not* a stationary point (gradient norm about 105, almost all of it in ϕ), which inflates the raw curvature by roughly a factor of seven, so the code first descends to the basin minimum and takes the Hessian there. The measured scales are 0.0023–0.0030 for bonds and 0.032–0.071 for angles. With this whitening the maximum curvature is about 80, so $\Delta t = 10^{-3}$ gives $\Delta t\, H \approx 0.08$ and a single timestep is comfortable for all 60 modes.

The computational box. The box of (13) is replaced by a torus-aware variant. Torsion slots *wrap* to $(-\pi, \pi]$ rather than clip, which is exact because U_{eff} is 2π-periodic in every torsion. Wrapping is applied in physical units, because in whitened units a torsion period is $2\pi/D_{ii}$ and only ϕ, ψ have $D_{ii} = 1$. The containment diagnostic reports torsions as always inside, so ordinary wrapping is never miscounted as a boundary projection in Table 4-style accounting. Bonds and angles do carry a genuine boundary, and it is physical rather than an overflow guard: (50) contains $\log b$ and $\log \sin a$, so the box must enforce $b > 0$ and $a \in (0, \pi)$. The limits are $b \in [0.3, 3] \times b_{\text{ref}}$ and $a \in [0.10, \pi - 0.10]$ rad. They lie many thermal standard deviations from the sampled region and do not bind in normal sampling.

Reporting windows and the branch cut. ϕ is reported on the conventional window $(-180°, 180°]$: only 0.47% of the reference mass lies within 0.15 rad of that seam, so Euclidean metrics on ϕ are not materially distorted. ψ is not. The C5/β region genuinely wraps around $\psi = \pm180°$ and carries 8.2% of the mass there, which would make Euclidean W_2, MMD and FES treat $\psi = +179°$ and $\psi = -179°$ as maximally distant. We therefore report ψ on a window whose branch cut was *searched* for the lowest-density angle: the cut sits at $\psi = -20°$ (measured mass within 0.15 rad: 1.7×10^{-4}), giving the fundamental domain $(-20°, 340°]$. This is a choice of fundamental domain internal to E5; the shared metric code of Appendix C is untouched and Euclidean distance remains valid on the chosen domain, while basin labels use a torus metric independently. The caveat must be stated: an E5 Ramachandran plot is consequently *not* the conventional one—the α_R and C7ax features appear near $\psi \approx +290°$—and a ψ time series shows an apparent discontinuity at the cut. Ensemble metrics and basin labels are unaffected.

G.5 The reference

Well-tempered metadynamics. Unlike the four main systems, this target admits no grid, no exact mixture draw, and no reliable Laplace-mixture importance proposal, so the reference is itself a simulation. *Metadynamics* works as follows: one runs ordinary molecular dynamics while progressively adding to the potential a bias V_{bias} built as a sum of Gaussian "hills" deposited at the currently visited value of the collective variables. The accumulating bias fills in the free-energy wells that the trajectory has already visited, and the trajectory is thereby pushed out of them and explores the remaining ones. In the *well-tempered* variant the hill height decays with the bias already present, so the bias converges rather than diverging, and at convergence $V_{\text{bias}} = -\frac{\gamma-1}{\gamma}F$ with bias factor γ.

The settings are: two `CustomTorsionForce` collective variables (ϕ, ψ); `LangevinMiddle` integrator at 300 K with a 2 fs timestep; bias factor $\gamma = 8$; initial hill height 1.2 kJ/mol; hill width 0.35 rad; a 100×100 grid; deposition every 1000 steps. The protocol has two phases. In the first, the bias is grown. In the second, the bias is *frozen* and a production trajectory samples the biased ensemble.

The frozen-bias reweighting identity. Each production frame at collective-variable value s is reweighted back to the unbiased Boltzmann measure by

$$w(s) = e^{+\beta V_{\text{bias}}(s)} = \exp\Big[-\beta \tfrac{\gamma-1}{\gamma} F(s)\Big]. \tag{54}$$

Because OpenMM returns exactly $F = -\frac{\gamma}{\gamma-1}V_{\text{bias}}$, the weight in (54) *is* the applied bias, and reweighting a static bias is unbiased regardless of whether the bias has converged. Metadynamics convergence therefore controls the *variance* of the reference, not its bias. Reweighted frames are mapped into whitened internal coordinates and form the reference conformer pool; where unweighted clouds are required (W_2, MMD) they are produced by sampling-importance resampling, and are labelled approximate exactly as in Appendix D.

Two silent traps. Both are recorded here because they are the kind of error that produces a plausible but wrong reference, and both are now guarded fail-closed. *(i)* OpenMM allocates the metadynamics grid with the collective variables *reversed*, so `getFreeEnergy()` is indexed $[\psi, \phi]$. When the two grid widths are equal the transpose is invisible dimensionally and silently mirrors the free-energy surface through the origin. An orientation check now correlates the empirical collective-variable histogram against $\exp(-\beta F/\gamma)$ for F and for F^{T} and aborts on disagreement. *(ii)* 1 ns = 10^6 fs. An earlier steps-per-nanosecond expression computed as $1000/\Delta t_{\text{fs}}$ ran the reference silently $1000\times$ short.

Basins are defined by barriers, not by minima. A local minimum of a *numerically estimated* free-energy surface is not automatically a metastable state, and treating it as one corrupts every occupancy statistic. Raw minimum finding on $\widehat{F}(\phi, \psi)$ returned *five* candidates. Two of them had escape barriers of about 0.00 and 0.19 $k_{\text{B}}T$; one of these lay at $\phi = +178°$, a shoulder of the β/C5 region seen across the ϕ seam. Keeping them would make basin occupancy meaningless—the Voronoi boundary between two minima with no barrier between them is arbitrary—and in fact it corrupted two separate diagnostics: the per-basin free energies became convergence-noisy, and the positive-ϕ island occupancy counted ordinary β-region configurations, so that even plain ULA appeared to cross the barrier. Merging any pair of candidates not separated by at least $1\,k_{\text{B}}T$ restores the classical three-state picture C5/β, C7eq, C7ax. The $1\,k_{\text{B}}T$ criterion is not tuned: the resulting partition is unchanged for every threshold from 0.05 to $2.0\,k_{\mathrm{B}}T$, a $40\times$ plateau, and first changes at $2.4\,k_{\mathrm{B}}T$, which is the physical C7eq–C5/β separation of $2.53\,k_{\mathrm{B}}T$.

G.6 Construction of the jump bank

This subsection is the operational core of E5. It states, in order, which displacements are proposed, in what space they live, how they are screened, and what survives.

Step 1: candidates. For every *ordered* pair of distinct basins (i, j) we take the torus-minimal displacement of the collective variables between the two basin minima,

$$(\Delta\phi, \Delta\psi)_{i\to j} = \mathrm{wrap}\big(m_j - m_i\big), \qquad \mathrm{wrap}(u) \in (-\pi, \pi]^2, \tag{55}$$

where m_i is the minimum of basin i. With three basins this gives $3 \times 2 = 6$ candidates, and both homotopy directions are present *by construction* as the $\pm$ pairs. Using the wrapped difference is what makes the bank take the cheap route of Appendix G.2: the C7ax$\leftrightarrow$C5/β candidates move ϕ by $\pm 151^\circ$, i.e. around $\phi = \pm 180^\circ$, and not through $\phi = 0$.

Step 2: the atoms are pure torsion displacements. Each candidate is promoted to a vector r_a in the full 60-dimensional whitened internal space that is nonzero *only* in the ϕ and ψ slots and *exactly* zero in the other 58 slots. This was verified numerically, including after the shell jitter of Step 4 is added. Purity is not cosmetic: it is exactly the hypothesis of (52), and hence the reason the score can be evaluated from force-field energies alone. Because $D_{ii} = 1$ on those two slots, $\|r_a\|$ is a genuine rotation magnitude in radians.

Step 3: screening, or the forbidden-chord drop rule. A candidate is admissible only if its chord stays in energetically reachable territory. For each candidate we draw states from its *source* basin (weighted by the reference pool) and evaluate the actual score exponent $\beta[U(q) - U(q - \theta r_a)]$ on the Gauss–Legendre θ nodes. A candidate is rejected if that exponent exceeds the cap $M_{\max} = 600$ of Appendix A for more than 10^{-3} of the (state, θ) samples—i.e. if its chord crosses a forbidden region of the Ramachandran surface. Every retained and every dropped atom is logged with its geography and the reason. Any pair of basins left unconnected would then be reached by relay through an intermediate basin, as in the Müller–Brown bank.

The measured outcome is itself the main methodological finding of E5: *six of six candidates are retained, none is dropped, and the maximum exponent encountered is about* 4 *against a cap of* 600*.* The obstruction that forced relay-only atoms for Müller–Brown and excluded the diagonals for ϕ^4 simply does not arise once the coordinates are internal—which is the quantitative counterpart of Appendix G.3.

Step 4: shell, weights and drift cap. Weights are uniform, $w_a = 1/6$; the lesson from the two higher-dimensional model systems is that skewing weights by target mass ratios backfires. The shell half-width follows the same rule as those systems, $h = 0.1 \min_a \|r_a\| = 0.1868$, and a draw from atom a is $r = r_a + \varrho\, u_a$ with $\varrho \sim \mathrm{Unif}(-h, h)$ and $u_a = r_a/\|r_a\|$, so the jitter stays inside the same (ϕ, ψ) plane and preserves torsion purity. The taming scale in (13) is $c = 2h = 0.3736$.

Step 5: which space the jumps live in. Stated explicitly, because it is easy to misread. The jumps are *not* Cartesian atom-space moves, and they are *not* a two-dimensional collective-variable model. They are displacements of the full 60-dimensional internal coordinate whose support happens to be confined to the two-dimensional (ϕ, ψ) subspace. The sampled object is the complete molecular configuration, the energy is the real force field evaluated on the reconstructed 22-atom geometry, and (ϕ, ψ) is only where we *measure*.

From	To	$\Delta\phi$ (deg)	$\Delta\psi$ (deg)	$\|r_a\|$ (rad)
C5/β	C7eq	+72.0	−79.2	1.868
C5/β	C7ax	−154.8	+136.8	3.606
C7eq	C5/β	−72.0	+79.2	1.868
C7eq	C7ax	+133.2	−144.0	3.424
C7ax	C5/β	+154.8	−136.8	3.606
C7ax	C7eq	−133.2	+144.0	3.424

Table 5: The six retained jump atoms, given by their nonzero (ϕ, ψ) components; all 58 remaining internal components are exactly zero. Weights are uniform (1/6). Norms are in radians because ϕ and ψ are unwhitened. All six candidates passed the forbidden-chord screen of Step 3 with maximum score exponent about 4 against the cap $M_{\max} = 600$.

The retained bank.

G.7 Estimator and residual certificate

Estimator. E5 deploys the paired multi-atom estimator LSC-CP-MA of Appendix A.1 and validates the exact arm *offline* instead of deploying it. The exact arm integrates the score of the full six-atom bank by deterministic product quadrature, $q_\theta \times A \times q_\rho = 16 \times 6 \times 8 = 768$ chord energies per particle per step. Nothing about a black-box force field forbids this—the chord identity needs energies and nothing else—but each of those 768 points expands into a full 22-atom reconstruction. Measured at the production ensemble, the exact arm runs at 1.67 s/step at its most favourable block size, with a 66 GiB peak; a sixteenfold smaller block costs only 16% more, so the arm is bound by arithmetic and memory traffic rather than by kernel-launch overhead and cannot be tuned out of its cost. That is about 18.5 h over the 40,000-step horizon against about 3 h for the realized-measure arm. This ratio is not an obstruction of principle but it is the practical point E5 exists to make: on a real force field the exact score is affordable as a *check* and not as a sampler.

Per particle and per step a bank $\{R_{n,a}\}$ is drawn from the shell laws; the score is the exact score of that realized finite measure, evaluated on Gauss–Legendre θ nodes; and the jump is $\sum_a N_{n,a} R_{n,a}$ with $N_{n,a} \sim \text{Poisson}(\lambda w_a \Delta t)$ conditional on the same realized measure. Score and jumps thus use one and the same measure, so the chord cancellation (8) holds atom by atom up to the chord quadrature.

Certificate. The weak-identity residual is evaluated in the *shifted*, atomwise form, as for the 24-dimensional ϕ^4 system. Substituting $q \mapsto q + \theta R_a$ in the score term of (8) converts the exponentially weighted integrand into a plain μ-expectation, and the atomwise residual for a test function f is

$$\mathcal{R}_a(f) = \Big| \int_0^1 \mathbb{E}_\mu\big[R_a \cdot \nabla f(q + \theta R_a)\big]\, \mathrm{d}\theta - \mathbb{E}_\mu\big[f(q + R_a) - f(q)\big] \Big|, \tag{56}$$

with the θ integral replaced by the same Gauss–Legendre rule used at run time and both expectations taken against the reweighted metadynamics pool. Two restrictions on f are essential and are worth stating as design rules. First, test functions must be *periodic* on the torus, so we use

$f(q) = \sin(m\phi + n\psi + c)$; a tanh-ridge test function of the kind natural on $\mathbb{R}^d$ is inadmissible here because it is not a function on the state space. Second, they must be *jump aligned*: a random ridge direction in $d = 60$ has overlap $O(d^{-1/2})$ with a jump atom and is therefore nearly blind to the quantity being certified. Measured values are $\mathcal{R} = 4.9 \times 10^{-14}$ on the generous full-torus domain and 3.6×10^{-15} on a tight single-basin domain.

The unshifted (direct) form, in which the exponentially weighted score field is integrated against μ directly, is computed and reported—its value is about 8.5×10^3 on the full torus and 1.4×10^1 on the tight domain—but it is *not* used as a gate. Its integrand $\mu\, S$ is $O(1)$ precisely where μ is exponentially small, so it is not μ-estimable in this dimension, and a large value of it is a statement about estimator variance rather than about stationarity. As in Appendix E, neither form tests the exponent cap, drift taming, Euler splitting, or box projection, and neither is a certificate of the invariant law of the finite-step chain.

Quadrature selection, and where E5 departs from E1–E4. The four main systems choose (q_θ, q_ρ) by refinement: they sweep a grid and take the smallest setting whose certificate residual is below 10^{-6} *and* whose terminal metrics agree with the finest setting's to within tolerance. E5 does not run that sweep; it pins the quadrature at the common default $(q_\theta, q_\rho) = (16, 8)$. Two reasons and one caveat, stated here rather than left to be inferred from the tables. First, the residual at the default is $\approx 5 \times 10^{-14}$, eight orders below the gate, so the first criterion is nowhere near binding. Second, the deployed estimator is the paired multi-atom score, which consumes q_θ only: q_ρ parameterizes the ρ-quadrature of the *continuous* shell law and is not a parameter of the realized-measure estimator at all, so two of the three axes E1–E4 sweep do not exist here. The caveat is that refinement's second criterion—terminal metrics converged against a finer quadrature—is correspondingly *not* tested for E5, and the reported quadrature is therefore justified by the certificate and by the estimator's structure rather than by a convergence sweep.

G.8 Cross-validation of the realized-measure estimator

Because the exact arm is not deployed here, the substitution has to be paid for with evidence rather than asserted. Two independent claims are checked, both at the production configuration (40,000 steps, 1000 particles per seed).

(A) Pointwise unbiasedness. The sharp statement is that averaging the realized-measure score over its bank draw returns the exact score, $\mathbb{E}_{\text{bank}}[S_{\text{MA}}(x)] \to S_{\text{exact}}(x)$ at fixed x. Over 4096 bank draws at 256 states sampled from the reweighted reference pool, the relative error against the exact score has median 2.3×10^{-4} and ninetieth percentile 1.6×10^{-3}, and the ϕ-components correlate at $1 - 4 \times 10^{-9}$. This is a statement about the estimator alone and is independent of any dynamics.

(B) End-to-end agreement. Pointwise unbiasedness need not survive 40,000 steps of feedback between score and state, so the two arms are also run as full samplers on a reduced two-seed set. Against $p^\star = (0.5697, 0.4220, 0.0083)$ the exact arm reaches $\widehat{p} = (0.567, 0.4195, 0.0135)$ and the realized-measure arm $\widehat{p} = (0.5825, 0.4065, 0.0110)$, i.e. basin L_1 errors of 0.0104 and 0.031; neither produced a non-finite value. Both recover the sparse island to the same order, with the exact arm the more accurate of the two, which is the expected direction: the realized-measure arm pays a variance price for not enumerating the bank. The gap is the honest cost of the substitution, and it is small against the differences between method families reported below.

G.9 Setup summary and validation ladder

Quantity	Value
molecule	alanine dipeptide (Ac-Ala-NHMe), vacuum, 22 atoms, fully flexible
force-field terms	21 bonds, 36 angles, 52 torsions, nonbonded with 98 exceptions
Cartesian dimension	$3N = 66$
sampled dimension d	60 (21 bonds, 20 angles, 19 torsions)
temperature	$T = 300\,\mathrm{K}$; $k_{\mathrm{B}}T = 2.4943$ kJ/mol; $\beta = 0.40091$ mol/kJ
collective variables	$\phi = q_5$, $\psi = q_8$ (backbone torsions)
whitening	diagonal, fixed; $D_{ii} = 1$ on ϕ, ψ; else $\min\{1, (\beta H_{ii})^{-1/2}\}$
box	torsions wrap to $(-\pi, \pi]$; $b \in [0.3, 3]b_{\mathrm{ref}}$; $a \in [0.10, \pi - 0.10]$
methods	ULA, MALA, FLA, BAOAB, PT, raw CP, LSC-CP-MA
particles N / seeds	1000 / 16
horizon T / Δt / steps	40 / 10^{-3} / 40,000
initialization	C7eq
jump rate λ	1
bank	$A = 6$ pure-torsion atoms (Table 5), weights 1/6
shell half-width / drift cap	$h = 0.1 \min_a \|r_a\| = 0.1868$ / $c = 2h = 0.3736$
score estimator	MA (paired multi-atom); exact (768 chords) cross-validated offline
exponent cap	$M_{\max} = 600$
reference	well-tempered metadynamics, $\gamma = 8$, frozen-bias reweighting

Table 6: E5 configuration. Metrics are the shared suite metrics of Appendix C, unchanged.

Check	Measured	Declared threshold
torch force field vs. OpenMM energy (1024 configurations)	$\approx 10^{-15}$ rel.	10^{-6}
torch force field vs. OpenMM force (1024 configurations)	$\approx 10^{-15}$ rel.	10^{-5}
analytic gradient vs. central differences	2.4×10^{-9}	—
BAT round trip $x \to q \to x$	2×10^{-15}	—
analytic $\log\lvert\det J\rvert$ vs. numerical log-determinant	1.4×10^{-14}	—
torsion derivative of $\log\lvert\det J\rvert$	exactly 0.0	—
Jacobian-free chord identity (52)	4.6×10^{-13}	—
shifted atomwise residual (56), full torus	4.9×10^{-14}	—
shifted atomwise residual, single basin	3.6×10^{-15}	—

Table 7: Validation ladder for E5. Each row is a component check of the implementation; together they establish that the energy, the coordinate transform, its Jacobian, and the chord cancellation are correct to the stated accuracy. They do not address the finite-step modifications listed in Appendix A.1.

G.10 Results

Table 8 reports terminal metrics over sixteen seeds, and the picture is the same one the four synthetic systems draw, now on a real force field. The slow event is reaching and correctly weighting the sparse positive-ϕ island C7ax, whose reference mass is only $\widehat{p}^{\star}_{\mathrm{C7ax}} = 0.0083$. Raw CP, which applies the jumps with no score correction, is badly biased there: it converges to its own stationary law and over-populates the island by more than forty-fold ($\widehat{p}_{\mathrm{C7ax}} = 0.39$), the largest basin error in the table. The Lévy-score correction removes exactly this bias: LSC-CP-MA recovers $\widehat{p}_{\mathrm{C7ax}} = 0.0096$ against the target 0.0083, a basin Kullback–Leibler of 1.1×10^{-3}, matching the gold-standard nonlocal method PT (0.0088, 6.5×10^{-4}) and improving on it in sampling efficiency: LSC-CP-MA attains the highest worst-basin effective sample size in the run, 1586 against PT's 1010.

The purely local integrators tell the complementary half of the story. ULA, MALA and BAOAB equilibrate the two dominant negative-ϕ basins (which are separated by a crossable $\sim 2.4\,k_{\mathrm{B}}T$ barrier) and so post small basin total variations, but they essentially never reach the island across the horizon ($\widehat{p}_{\mathrm{C7ax}} \leq 4 \times 10^{-4}$, and a worst-basin ESS of 0 for BAOAB): their low aggregate error is carried entirely by the 99% of mass in the two easy basins. FLA, whose invariant law is not π, overweights the positive-ϕ region ($\widehat{p}_{\mathrm{C7ax}} = 0.26$) and is recorded but not a π-targeting comparator. The free-energy-surface RMSE is a weak discriminator here (1.2–1.4 $k_{\mathrm{B}}T$ for every π-targeting method, dominated by the sharp basin walls rather than the rare-island mass); the basin occupancy and its KL are what separate the methods. The exact deterministic-quadrature arm is not deployed in this table; its offline cross-validation against LSC-CP-MA is reported in Appendix G.8.

Method	$\mathrm{TV}_{\mathrm{basin}}$	$D_{\mathrm{KL}}(p^\star \\| \widehat{p})$	$\mathrm{RMSE}_F/k_{\mathrm{B}}T$	$\widehat{p}_{\mathrm{C7ax}}$	W_2
ULA	0.0258 (0.013)	0.0133 (0.0047)	1.35 (0.039)	0.0004	0.121 (0.02)
MALA	0.0177 (0.011)	0.0146 (0.0033)	1.36 (0.031)	0.0002	0.120 (0.016)
FLA	0.250 (0.014)	0.261 (0.018)	1.59 (0.024)	0.2584	1.32 (0.042)
BAOAB	0.0178 (0.0084)	0.0161 (5.3×10^{-4})	1.37 (0.018)	0.0000	0.116 (0.013)
PT	0.0117 (0.0079)	6.5×10^{-4} (5.5×10^{-4})	1.23 (0.048)	0.0088	0.153 (0.034)
raw CP	0.378 (0.012)	0.446 (0.020)	1.72 (0.038)	0.3866	1.62 (0.033)
LSC-CP-MA	0.0136 (0.010)	1.08×10^{-3} (9.9×10^{-4})	1.22 (0.046)	0.0096	0.176 (0.041)

Table 8: Terminal E5 metrics at $t = T$, mean (std) over seeds, against the reweighted metadynamics reference. $\mathrm{TV}_{\mathrm{basin}}$ and D_{KL} use the three barrier-merged basins C5/β, C7eq, C7ax; RMSE_F is the additive-aligned free-energy-surface RMSE on the (ϕ, ψ) grid in the reporting windows of Appendix G.4; $\widehat{p}_{\mathrm{C7ax}}$ is the recovered mass of the positive-ϕ island; W_2 is the sliced Wasserstein-2 distance in whitened internal coordinates. Metric definitions in Appendix C.

G.11 Limitations

Three limitations are specific to E5 and are stated before any numbers are read.

The reference is an experimental input, not ground truth. Unlike the double well, MoG40 and Müller–Brown, whose references are grid or exact draws, the E5 reference is a finite metadynamics simulation. The frozen-bias identity (54) makes reweighting unbiased for any static bias, so incomplete metadynamics convergence inflates the *variance* of the reference rather than displacing it; but the reference remains convergence-limited, and, as with the ϕ^4 importance reference of Appendix D, differences below its own uncertainty are reference sensitive.

The island mass is the least precisely known quantity in the experiment. C7ax carries under 1% of the target probability, so its reference mass is estimated from the smallest sub-population of the reweighted pool. Its uncertainty must be quoted with it, and comparisons of $\widehat{p}_{\mathrm{C7ax}}$ across methods should be read against that uncertainty and not against the point value alone.

Scope. E5 is a single molecule, in vacuum, at one temperature, with a bank built from three known basins of a two-dimensional collective-variable description. It demonstrates that the construction survives contact with a real force field once the coordinates are chosen so that a conformational transition is an affine displacement. It is not a molecular benchmark, it makes no claim about solvated systems, larger peptides, or unknown basin structure, and, as everywhere in this note, its passage statistics would be algorithmic mixing diagnostics rather than physical kinetics.